

Integrating High-Resolution LiDAR Data with Cellular Automata for Wildfire Spread Prediction: Development, Calibration, and Validation

by

Andreas Paphitis

University of Amsterdam
Faculty of Science
Graduate School of Life and Earth Sciences

Supervisor: dr. A.C. Seijmonsbergen
Assessor: dr. Yifang Shi

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ABSTRACT

This study develops a cellular automata (CA) wildfire model enhanced with high-resolution LiDAR data to simulate fire spread in Tenerife’s *Pinus canariensis* forests, using the 2023 wildfire as a case study. The model integrates detailed 3D fuel structures derived from LiDAR point clouds through systematic preprocessing, capturing surface, canopy, and ember-driven fire dynamics in complex volcanic terrain. A terrain preprocessing pipeline combines pre-computed terrain features with dynamic wind calculations, enabling efficient simulation of fire behavior in barrancos and mountainous landscapes. The model successfully incorporates LiDAR data and runs efficiently, but validation reveals unrealistic, overly aggressive spread patterns due to insufficient fire suppression mechanisms and fuel constraints. These findings highlight the need for refined CA rules that ensure ecological realism, offering critical insights for improving fire prediction and management strategies in Mediterranean ecosystems.

1 Introduction

Wildfires are a complex ecological phenomenon that are both a natural process vital to ecosystem health and a potential hazard to both human and ecological communities (Alcasena et al., 2021; Gajendiran et al., 2024). Their occurrence, intensity, and behavior are based on a wide range of factors including climate, vegetation, topography, as well as human activity. In recent years, the frequency and severity of wildfires have intensified due to changing climatic patterns, land use shifts, and historical fire suppression practices (Kreider et al., 2024). This has challenged traditional wildfire management, creating the need advanced tools for use in decision-making and management efforts. This thesis addresses this need by developing a spatially explicit wildfire model based on Cellular Automata (CA), enhanced with LiDAR-derived data to capture vertical fuel structures, with a focus on fire-prone *Pinus canariensis* (*P.c*) forest ecosystems. By integrating ecological principles, high-resolution LiDAR data, and computational methods, this research aims to improve understanding of wildfire dynamics and support effective risk management in Tenerife and similar regions.

1.1 Background and Motivation

Changes in climatic patterns, especially in Mediterranean contexts, favor prolonged

drought periods and increased temperatures, leading to increased fire risk and longer fire seasons that alter natural fire cycles (Carrillo et al., 2022). Furthermore, rapid expansion of the wildland-urban interface (WUI)—which grew 41% in housing units and 33% in land area between 1990-2010 in the United States (Radeloff et al., 2018), has enhanced wildfire hazards by increasing both ignition sources and community exposure.

Associated with urban expansion are changes in fire management practices. Historically, fire suppression policies aimed to protect communities by rapidly subduing wildfires or managing fuel loads through mechanical or chemical means (North et al., 2015). However, these practices have produced unintended ecological consequences, including fuel accumulation (leaf litter, deadwood, understory vegetation) that increases fire severity (Stephens et al., 2013). The suppression of moderate fires causes larger fuel buildups; when fires exceed anthropogenic suppression capacity, they become high-severity wildfires that threaten both ecosystems and human communities (Kreider et al., 2024).

Fire modeling has become essential for predicting fire behavior, as well as supporting management decisions, in regions where climate change, land-use shifts, and suppression policies have compounded fire risks. While changing management practices—such as al-

lowing more low-intensity fires or implementing prescribed burns—offers solutions to restore ecological balance and reduce fuel loads, this alone is insufficient. In an era of increasing droughts, rapid WUI expansion, and unpredictable weather, even controlled burns could exacerbate risks without advanced modeling to inform safer decisions (Hessburg et al., 2021). An integrated approach combining proactive management with advanced modeling enables data-driven decisions that balance restoration with protection, as evidenced by frameworks optimizing prescribed fire use and mitigating wildfire impacts (Kalabokidis et al., 2016)

Fire modeling has evolved from simple conceptual tools and empirical approaches like Rothermel’s surface fire spread equations (Rothermel, 1972) to physics-based simulations. Rothermel’s approach, used in systems like BEHAVE, offers fast predictions but depends on historical data, limiting accuracy in extreme conditions (Andrews P. L., 1986). Physics-based models like FIRETEC provide detailed simulations but incur prohibitive computational costs for real-time decisions (Coen et al., 2013). These limitations—particularly struggles with spatial heterogeneity, stochastic behavior, and large-scale applications—have driven development of simulation-based approaches integrating discrete spatial rules for scalable predictions (Sullivan, 2009).

Among such approaches, cellular automata (CA) offer practical, scalable methods for simulating fire spread across diverse landscapes. CA models use local rules based on slope, wind, and fuel to determine ignition, producing realistic large-scale fire patterns (Karafyllidis & Thanailakis, 1997). Their computational efficiency makes them particularly valuable for operational applications: CA models predict fire progression under varying conditions to accurately determine fire perimeters for actions such as timely population evacuation, optimize resource deployment (firefighting crews, aerial support), and evaluate long-term impacts of land-use changes on ignition probabilities (Perello et al., 2024). Compared to machine learning or physics-based alternatives, CA models demonstrate faster calibration

(hours vs. days-weeks) and shorter run-times (Pais et al., 2021). Studies have shown then landscape-level CA simulations complete in minutes—up to $30\times$ faster than vector-based models like Prometheus—making them suitable for long-term planning, post-event analysis, and near-real-time emergency support (Pais et al., 2021). CA calibration leverages existing empirical systems and are often achievable in hours using limited data, contrasting with ML’s extensive training periods on large historical datasets or physics-based models’ reliance on detailed laboratory parameters demanding intensive computation (Pais et al., 2021). CA adaptable local rules reduce reliance on outdated historical data, particularly valuable for rapidly evolving fire regimes where past conditions may not represent current risks amid shifting climates.

However, traditional CA models often overlook alternative fire spread modes, including vertical fire dynamics—the upward spread from surface fuels to canopy via ladder fuels—and spotting ignition caused by firebrands, which trigger intense crown fires and rapid expansion (Koo et al., 2010; Liu et al., 2018). This limitation was apparent during the 2023 Tenerife fire (nearly 12,000 hectares burned), where vertical spread and long-distance spark projection contributed to rapid escalation that existing models failed to predict accurately.

Deficits were identified in advanced fire prediction tools, including data scarcity, complex fire-induced microclimates (thermal inversions), and non-linear propagation patterns (Correa et al., 2025). The fire’s classification as "sixth-generation" wildfire—extreme intensity, rapid propagation, long-distance spotting, and self-generated atmospheric dynamics—led to under-performance of operational prediction tools (Correa et al., 2025). Post-fire analysis revealed conventional models emphasizing horizontal spread rates and surface fire behavior failed to forecast progression within topographically complex contexts, leading to greater reliance on empirical analysis and territorial knowledge over simulation-based tools (Correa et al., 2025).

The fire’s transition from surface to crown

burning, enabled by continuous fuel layers (dense shrub thickets, flammable undergrowth) combined with wind-driven firebrands, contributed to massive escalation and self-generated meteorological conditions that complicated resource allocation and evacuation efforts across multiple municipalities (Correa et al., 2025). High-severity burns covered approximately 50% of the affected area—particularly impacting the 80% wooded forest component (Correa et al., 2025). This aligns with broader research showing that accumulated biomass from rural abandonment could have facilitated upward fire movement into canopies and spot fires ahead of the main perimeter, exacerbating crown fire dynamics in pine-dominated ecosystems (Fernández-García et al., 2019; Keeley, 2012).

The need for improved CA models is further emphasized by requirements for multiple evaluation metrics assessing model performance across fire behavior dimensions. Traditional single-metric approaches (total burned area comparison) fail to capture complexity of fire spread patterns, temporal progression, and spatial accuracy simultaneously. Integrated evaluation frameworks incorporating spatial similarity metrics, temporal progression analysis, and fire spread characteristics enable more robust validation and calibration. These multiple metrics are particularly crucial when validating against real fire events like the 2023 Tenerife fire, where different aspects (horizontal spread, vertical transitions, ember transport) contributed to overall progression and should be evaluated independently.

Developing enhanced CA models incorporating vertical fire connectivity, spotting ignition, and multi-dimensional spread mechanisms is therefore needed to account for fuel structure and fire propagation complexity driving unpredictable wildfire behavior. However, the gap between model capability and validation against observed fire behavior represents a critical knowledge deficit limiting operational deployment of fire prediction systems. The 2023 Tenerife fire provides an ideal case study, as detailed EMSR delineations captured fire progression across multiple days, enabling validation of

both spatial accuracy and temporal dynamics. This approach facilitates creation of validated tools informing fire management decisions and emergency response planning in Mediterranean fire-prone landscapes.

1.2 Objectives and Research Questions

Building on the above context, this research improves wildfire spread modeling for Tenerife’s pine forests by integrating LiDAR-derived vegetation structure with cellular automata approaches. The main goal is to develop and validate a modeling framework that captures multi-dimensional fire behavior in complex volcanic terrain. To achieve this, we address two specific objectives and research questions rooted in identified gaps:

- **Objective 1:** Integrate LiDAR-derived forest structure into cellular automata wildfire modeling to represent vertical fuel continuity and enable simulation of multi-dimensional fire spread mechanisms.
- **Research Question 1:** How effectively can LiDAR-derived Plant Area Density (PAD) data represent vertical fuel connectivity in cellular automata models, and to what extent does this improve simulation of crown fire behavior compared to traditional two-dimensional approaches?

This objective addresses the gap in representing vertical fuel continuity that contributed to model failures during the 2023 Tenerife fire, where crown fire development was underestimated by conventional models.

- **Objective 2:** Develop and validate a framework incorporating multiple fire spread modes (horizontal, vertical, ember transport) and quantify their relative contributions to fire progression in complex volcanic terrain.
- **Research Question 2:** How do different fire spread modes (horizontal surface spread, vertical fire spread, and long-distance ember spotting) contribute to

overall fire behavior in Tenerife’s topographically complex pine forests?

This objective addresses the documented underestimation of vertical spread and ember transport mechanisms during the 2023 Tenerife fire, requiring integrated evaluation metrics to assess each spread mode’s contribution independently.

2 Changing Fire Regimes in Canary Island Pine Forests

The Canary Island pine forests of Tenerife exemplify complex fire regime dynamics shaped by natural fire cycles, human intervention, and climate change, providing context for contemporary wildfire modeling challenges.

2.1 Historical Fire Dynamics and Human Influence

Early Spanish colonial records describe extensive forest exploitation and laws against setting fires, indicating anthropogenic burning concerns (Molina-Terrén et al., 2016). By the 19th and early 20th centuries, human-started fires had become commonplace. Tree-ring fire scar studies on Tenerife and La Palma showed fire return intervals of 1.4-1.7 years for all fires and 11.2-15.2 years for larger fires during 1850-1960 (Molina-Terrén et al., 2016). These frequent surface fires were largely attributed to land-use practices (pastoral fires, agricultural burns) rather than natural lightning ignitions (Molina-Terrén et al., 2016).

In this pre-suppression era, fires were frequent but generally low to moderate severity, allowing ecosystem adaptation. Dendrochronological records from 1850-2007 on Tenerife, La Palma, and El Hierro indicate many pine stands burned every few years prior to 1960, with pines often surviving multiple fires as evidenced by repeated fire scars (Molina-Terrén et al., 2016). However, after 1960, fire suppression policies and reduced fuel collection practices disrupted this pattern, leading to reduced frequency but higher intensity fires (Molina-Terrén et al., 2016).

2.2 Ecological Role and Fire Adaptation of *Pinus canariensis*

P. c possesses exceptional fire adaptations including thick insulating bark, epicormic resprouting capability even after crown scorch, and partial serotiny with post-fire seed release (Climent et al., 2004). These traits make Canary Island pine among the world’s most fire-resistant pines, enabling survival of high-intensity crown fires that would devastate most coniferous forests. In healthy fire regimes with occasional low-to-moderate severity fires, stands remain dominated by old surviving pines while younger trees regenerate in canopy openings—a fire-maintained age structure historically typical of these forests (Molina-Terrén et al., 2016).

Mild fires historically maintained ecosystem balance by consuming litter and shrubs, recycling nutrients, creating seedbeds for regeneration, and enhancing understory diversity compared to mechanical fuel removal (Arévalo et al., 2023). High-intensity fires can adversely affect soil conditions (Durán et al., 2019), but fire adaptations enable recovery: serotinous cones disperse seeds onto ash-fertilized soil while epicormic resprouting allows mature pine survival (Climent et al., 2004). This dual strategy supports forest rejuvenation, enabling infrequent moderate-severity fires to promote new growth without eliminating established trees. The historical regime of frequent anthropogenic fires before 1960 maintained a resilient system; the transition to less frequent but larger fires after 1960 altered these dynamics.

2.3 Contemporary Changes and Climate Impacts

Around 1960, as tourism grew and modern forestry practices took hold, authorities implemented aggressive fire suppression policies (Molina-Terrén et al., 2016). Fire history analyses show a clear regime shift: "After ca. 1960, the fire regime changed to a more widespread pattern at a lower frequency" (Molina-Terrén et al., 2016)—replacing many small fires with

few very large fires. Critically, post-1960 widespread fire years correspond to abnormally dry conditions, whereas earlier fires showed no strong climate trigger (Molina-Terrén et al., 2016). By the 2000s, decades of suppression meant escaped fires tended toward high-intensity crown fires under hot, dry weather, with "fire suppression essentially eliminating] all but the largest, higher-intensity wildfires, establishing a new fire regime" (Molina-Terrén et al., 2016).

Land-use changes compounded these effects. Traditional agro-pastoral activities (livestock grazing, controlled burns) that maintained low fuel loads have been largely abandoned in Tenerife’s highlands (Molina-Terrén et al., 2016). Pine forest regeneration on former terraces and pastures increased forest cover but created more continuous vegetation capable of carrying fire.

Climate projections indicate these trends will intensify. Globally, the fire weather season increased 18.7% between 1979-2013, with over 25% of vegetated land experiencing significant extensions (Jolly et al., 2015). For the Canary Islands specifically, Carrillo et al. (2022) project that by century’s end, the annual fire season could lengthen by up to 75 days, with extreme fire weather days increasing 58% and a 44% expansion in high-risk areas—particularly at higher altitudes where climate sensitivity is most pronounced.

3 Methodology

Area Description Tenerife, the largest of Spain’s Canary Islands (2,034 km²), is situated approximately 300 kilometers west of the African coast. The island’s topography ranges from sea level to Mount Teide, at 3,718 meters, creating diverse microclimates and ecological zones supporting over 800 endemic species (Del Arco Aguilar & Rodríguez Delgado, 2018). Canary Island pine (*Pinus canariensis*) forests dominate elevations between 1,000-2,200 meters.

On August 15, 2023, a catastrophic wildfire erupted in Arafo municipality, rapidly expanding across 12 municipalities. Declared extin-

guished on November 10, 2023, the fire scorched approximately 12,000 hectares—the most severe wildfire in the Canary Islands in four decades and in Spain in 2023 (Center, 2023). Rugged volcanic terrain, including steep barrancos (volcanic ravines), and dense vegetation accelerated spread and complicated suppression efforts, requiring evacuation of over 12,000 residents.

This event exemplifies the need for sophisticated modeling approaches in complex insular environments where traditional fire prediction tools underperformed. The 2023 fire provides an ideal case study due to documented fire progression through EMSR-685 rapid mapping across multiple days, enabling model calibration and validation against observed fire behavior in Mediterranean pine ecosystems.

Data Sources LiDAR’s precision in mapping 3D fuel structure deals with the limitations of more traditional satellite-based imagery (Kramer et al., 2014). For example, such detailed data can detect changes in vertical fuel continuity after treatments, though dense canopies may occlude understory layers (Olzewski & Bailey, 2022). Despite this, LiDAR remains one of the most effective tools for landscape-scale fuel mapping, enabling simulation of vertical fire dynamics.

This study integrates three primary data sources for fire simulation modeling:

LiDAR Data . High-resolution LiDAR datasets for Tenerife, Canary Islands, Spain, obtained from the Plan Nacional de Ortofotografía Aérea (PNOA) under the Instituto Geográfico Nacional (IGN) at 0.5 pts/m^2 from the 2016 coverage (Appendix A2). These data provide detailed 3D vegetation structure information for fuel load estimation.

Digital Terrain Model (DTM) . DTM data were obtained from IGN as pre-processed rasters at 5 m spatial resolution, which were then merged in QGIS to create a seamless elevation surface. These data provide the topographic foundation for terrain-fire interaction modeling (Appendix A1).

Fire Perimeter Data: Observed fire progression boundaries were obtained from the European Forest Fire Information System (EFFIS) Emergency Management Service (EMSR) Copernicus rapid mapping activations for the August 2023 Tenerife fire event. EMSR-685 provided georeferenced fire perimeter delineations for 4 different dates (labeled as Days 1-4 for ease of use) of the fire progression, serving as calibration and validation targets.

All datasets use the ETRS89 / UTM Zone 28N coordinate system (EPSG:25828) and are spatially aligned. LiDAR and DTM data were pre-processed to extract model inputs, while EMSR delineations were processed to create target fire perimeters for model calibration and validation. See Appendices A1 and A2 for detailed preprocessing specifications.

3.1 LiDAR Data Preprocessing and Vegetation Structure Integration

A complete visual representation of the entire methodology workflow can be found in Appendix 5

The LiDAR point cloud was pre-processed to obtain ecologically meaningful vegetation metrics suitable for fire simulation modeling. This was done through a multi-stage pipeline including height normalization, vegetation structure quantification, and resolution optimization for computational efficiency, see Fig.1 and Table 1

LiDAR Processing Pipeline Overview

Height Normalization and Vegetation Classification Raw LiDAR point clouds contain absolute elevation values that must be normalized to Height Above Ground (HAG) in order to derive meaningful vegetation metrics. Height normalization was performed using PDAL’s `filters.hag_nn` algorithm, which applies nearest-neighbor interpolation to compute ground-relative heights for each point. This filter constructs a Triangulated Irregular Network (TIN) from ground-classified points and calculates vertical distances from non-ground returns to the interpolated surface, with extrap-

olation enabled to handle edge effects in complex terrain. Following normalization, vegetation points were isolated by filtering for ASPRS classification codes 3–5 (low, medium, and high vegetation). Negative height values we found what was a results from using nearest-neighbor interpolation in steep terrain. To address this, negative heights were excluded from NRD calculations while preserving non-negative values ≥ 0 m, consistent with standard methodologies for vegetation structure analysis (Campbell et al., 2018)

Vegetation Structure Quantification

Normalized Return Density (NRD) quantifies vertical vegetation distribution by measuring the proportion of LiDAR returns at each height layer relative to cumulative returns from ground level (Fig. 1C). This approach follows established methodologies in forest ecology and remote sensing literature (Campbell et al., 2018) . For height bin h :

$$\text{NRD}(h) = \frac{N(h)}{\sum_{i=0}^h N(i)} \quad (1)$$

The NRD formula is derived from standard remote sensing practices for vertical profiling, as it normalizes return densities to account for varying LiDAR penetration depths, ensuring comparability across heterogeneous vegetation (Campbell et al., 2018).

Plant Area Density (PAD) was derived from NRD using the Beer-Lambert law (Fig. 1D), providing physics-based vegetation density measures (Hosoi & Omasa, 2006; Riaño et al., 2003):

$$\text{PAD}(z) = \frac{-\ln(1 - \text{NRD}(z))}{\kappa \cdot \Delta z} \quad (2)$$

with extinction coefficient $\kappa = 0.6$ and height bin $\Delta z = 2$ m (detailed parameterization in Appendix A2.3). The Beer-Lambert law was selected for its established use in modeling light attenuation through vegetation canopies, directly linking LiDAR returns to biomass density (Hosoi & Omasa, 2006) . The value of $\kappa = 0.6$ assumes a random leaf angle distribution typical for pine and mixed forests, as validated in Mediterranean ecosystems (Riaño et

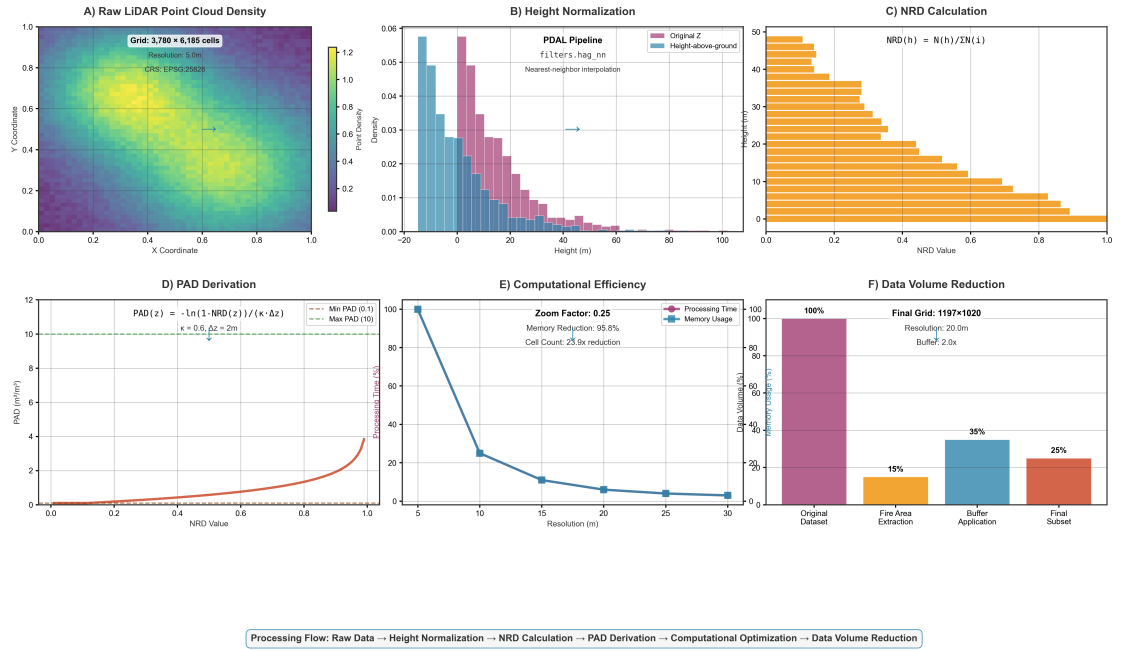


Figure 1: LiDAR Preprocessing Pipeline

Step	Processing Stage	Spatial Resolution	Reference
1	Height normalization (PDAL filters.hag_nn)	Point cloud	Appendix A2.1
2	ASPRS vegetation classification (classes 3-5)	Point cloud	—
3	Normalized Return Density (NRD) quantification	5 m	Appendix A2.2
4	Plant Area Density (PAD) via Beer-Lambert law	5 m	Appendix A2.3
5	Multi-scale resampling (three-framework interpolation)	5 m → 20 m	Appendix A2.4
6	Spatial subsetting to fire-affected areas	20 m	Appendix A2.5
7	Simulation integration and optimization	20 m	Appendix A2.6

Table 1: LiDAR Preprocessing Pipeline for Vegetation Structure Quantification

al., 2003) lower values (e.g., 0.5) were tested but increased underestimation of understory density. $\Delta z = 2$ balances vertical resolution with computational load, matching the LiDAR point density (0.5 pts/m²) and ecological scales of fuel

layering in Tenerife's pine forests.

3.1.1 Simulation Framework Integration

The preprocessed PAD data were integrated to the simulation framework with prioritizing computational efficiency in mind. PAD data from 20 vertical layers were resampled to 20 m resolution (1197×1020 cells) and stored as NumPy arrays. During simulation initialization, arrays were loaded and subsetting to simulation grid dimensions (609×609) using efficient array slicing (complete specifications in Appendix A2.6). This approach allowed for faster model initialisation while maintaining spatial accuracy. The resampling from 5 m to 20 m resolution reduces data volume by a factor of 16 while preserving key vegetation patterns, as finer resolutions (e.g., 5 m) increase computational time exponentially without proportional gains in fire spread accuracy for large-scale simulations (Sullivan, 2009).

3.2 Terrain Preprocessing Pipeline

This pipeline was used to process DTM into layers that affect fire behavior throughout the simulation. This stage removed computationally expensive runtime calculations by essentially pre-computing all terrain-fire interactions and storing them as arrays. The DTM is processed through a sequential workflow, each building upon previous outputs to create a comprehensive terrain representation for fire modeling (detailed algorithms in Appendix A1).

DEM Loading and Geographic Subsetting The DTM was loaded using rasterio in a Python environment and subsetting to the Day 4 EMSR delineation plus a 100% buffer zone (Appendix A1.1). All operations used UTM Zone 28N (EPSG:25828).

Slope and Aspect Calculation Slope and aspect are calculated using Sobel operators to detect terrain features (Appendix A1.2). Slope measures how steep the terrain is (in degrees), while aspect shows which direction the slope faces (0-360°). These control how fast fire

spreads uphill versus downhill in the simulation. Sobel operators were chosen for their efficiency in edge detection on raster data, providing robust gradient estimates with low computational overhead compared to finite difference methods (Sullivan, 2009)

Barranco Detection The model locates barrancos (volcanic ravines) by looking for areas that are both steep ($>25^\circ$ slope) and form natural drainage channels (Appendix A1.3). Barrancos are an important aspect of topography to consider because they act as corridors where fire spreads faster (Sullivan, 2009). The detection process creates three maps: where barrancos are located, where low-lying areas are found, and which direction water would flow through them.

3.2.1 Wind Effects Calculation

This calculates how terrain affects wind patterns (Appendix A1.4). Areas identified as barrancos get stronger wind ($2.5\times$ normal speed) because wind gets funneled through them. The calculated wind effects are saved as data files that the simulation can use directly.

3.2.2 Data Optimization and Storage

The terrain data is reduced from 5m resolution to 20m resolution, making it 16 times smaller while keeping the important terrain features (Appendix A1.5). All processed terrain data is saved as separate NumPy files with information about how the processing was done.

3.3 EMSR Data Preprocessing

Fire perimeter polygons from EMSR-685 were processed through a standardized workflow (Appendix A3). The original vector data were reprojected to EPSG:25828, then rasterized to match the simulation grid resolution (20m). Each daily fire perimeter was converted to binary masks indicating burned and unburned areas, see Fig. 2.

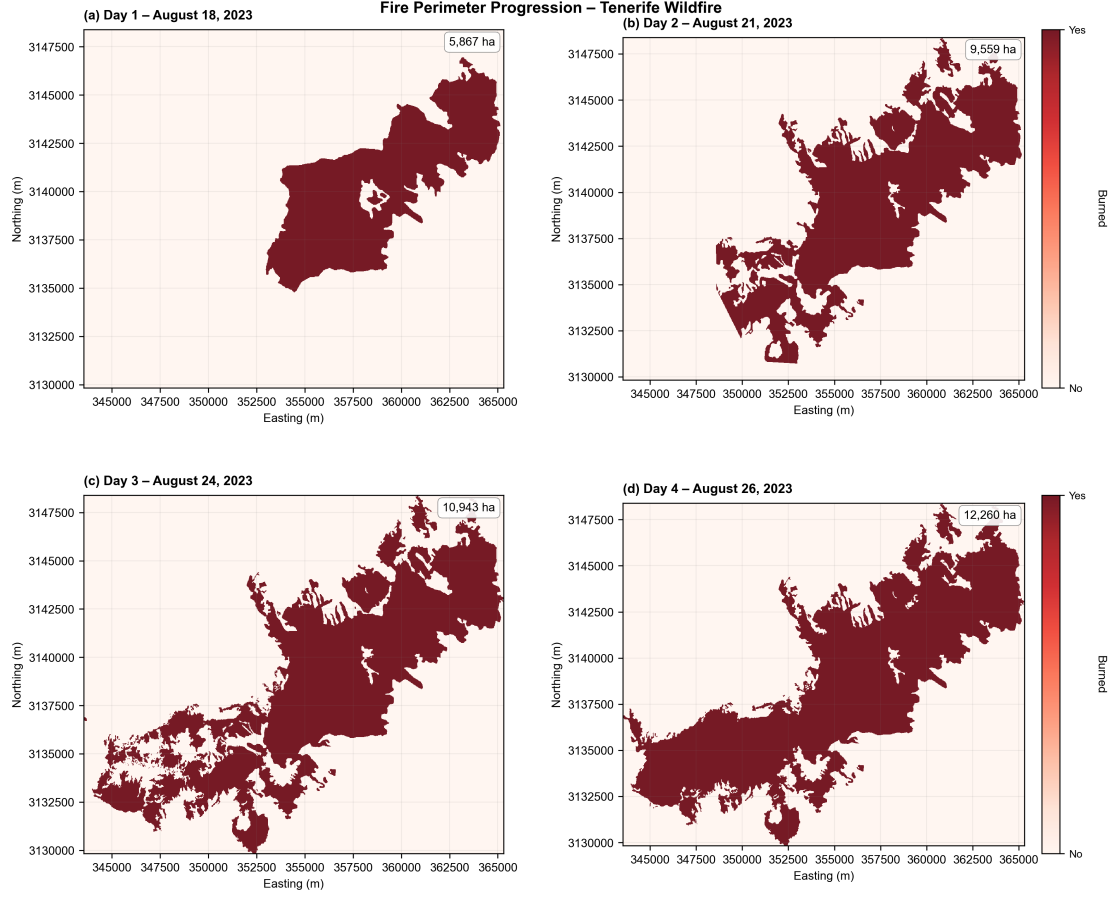


Figure 2: Binary masks showing fire perimeter evolution at 20 m resolution for (a) Day 1 – August 18, 2023 (5,867 ha); (b) Day 2 – August 21, 2023 (9,559 ha); (c) Day 3 – August 24, 2023 (10,943 ha); and (d) Day 4 – August 26, 2023 (12,260 ha). Days 1–2 were used for calibration; Days 3–4 for validation .

3.4 Study Area Selection

The simulation domain was defined based on the observed EMSR-685 Day 4 fire perimeter (12,260 ha), encompassing the fire-affected area plus a 10% buffer zone to capture edge effects and potential spread beyond observed boundaries. This resulted in a 148 km² domain (12.18 km × 12.18 km) at 20 m spatial resolution (609 × 609 grid cells), centered on the fire’s core spread area in Tenerife’s northeastern pine forests.

The domain was selected based on four criteria for fire behavior modeling studies:

1. **Diverse vegetation conditions:** Dense Canary Island pine forests with varying fuel loads captured through LiDAR (0.5 pts/m² point density), enabling representation of heterogeneous fuel structure
2. **Complex topography:** Elevation gradients (1,000-2,200 m), volcanic slopes (>40° in places), and barranco systems influencing wind channeling and fire spread patterns
3. **Documented fire progression:** EMSR-685 rapid mapping providing temporal snapshots (Days 1-4: August 18, 21, 24, 26) for calibration and validation

4. **Varied terrain features:** Slope, aspect, and elevation heterogeneity critical for testing terrain-fire interaction mechanisms

Data Extent Distinction: LiDAR-derived vegetation structure data were preprocessed over a larger extent ($24 \text{ km} \times 20 \text{ km}$, approximately 100% buffer beyond Day 4 perimeter) at 5 m resolution to ensure adequate spatial coverage for interpolation edge effects, then re-sampled to 20 m resolution and subset to the 609×609 simulation domain boundaries (10% buffer) for computational efficiency.

3.5 Model Architecture and Development

This study implements a custom three-dimensional cellular automaton (CA) fire spread model for three fundamental reasons.

First, LiDAR integration requirements. Widely-used operational fire models in Mediterranean contexts—including FARSITE, FlamMap, and Prometheus—operate on two-dimensional surfaces with discrete fuel classifications and cannot directly utilize continuous three-dimensional Plant Area Density (PAD) data from LiDAR (Scott & Burgan, 2005). Such models apply Rothermel’s surface fire spread equations and Van Wagner’s crown fire initiation criteria, which were developed for discrete fuel types rather than vertically continuous fuel profiles (Rothermel, 1972; Wagner, 1977). PAD data represent vertical fuel stratification, enabling representation of the ladder fuel dynamics and crown fire transitions observed during the 2023 Tenerife fire (alexander; Cruz et al., 2005; Sullivan, 2009).

The second consideration was computational efficiency. The study domain encompasses $1,197 \times 1,020$ cells at 20 m resolution across 20 vertical layers (24.4 million cells), with calibration requiring hundreds of simulation runs. The model uses sparse matrix storage and pre-computed terrain interactions—including baranco wind channeling effects—reducing memory requirements by approximately 85% compared to dense array storage. This optimization

enables calibration-oriented batch processing with hundreds of systematic parameter combinations within practical time allocations (Pavlidou, 2011).

Thirdly, region-specific calibration. A lot of standard fire behavior fuel models were developed primarily for North American ecosystems and do not capture Canary Island pine’s unique fire adaptations or regional atmospheric phenomena like calima events (Scott & Burgan, 2005). The custom model exposes all fire behavior parameters—spread probability, fuel consumption rate, ember generation, and terrain influence coefficients—for direct optimization against observed EMSR progression data through systematic sensitivity analysis and grid search calibration (Hamby, 1994).

The cellular automata model implements three distinct fire spread mechanisms operating simultaneously within the three-dimensional forest grid:

3.5.1 Horizontal Fire Spread

Horizontal fire spread follows CA principles with Moore neighborhood connectivity (8 adjacent cells). The model evaluates ignition probability for each adjacent cell using a multiplicative approach. Target cells require unburned state and fuel load above minimum threshold.

$$P_{\text{ignition}} = P_{\text{base}} \times F_{\text{fuel}} \times F_{\text{wind}} \times F_{\text{slope}} \times F_{\text{distance}} \quad (3)$$

This follows established wildland fire behavior modeling principles, where individual factors represent physical processes that work together to determine ignition success (Sullivan, 2009). The multiplicative approach ensures that ignition probability approaches zero when any critical factor becomes limiting (e.g., no fuel, extreme upwind conditions, steep downslope spread), consistent with observed fire behavior. This approach also naturally bounds the probability to physically meaningful ranges without requiring arbitrary constraints, as each factor typically ranges between 0.5 and 1.5. The base probability represents the intrinsic spread potential under neutral conditions (flat terrain, no wind, optimal fuel). For the Tener-

ife calibration, $P_{\text{base}} = 0.95$ was determined through focused grid search calibration against EMSR Day 1-2 fire perimeters, reflecting exceptional fire weather during the event.

Fuel Availability Factor

$$F_{\text{fuel}} = \begin{cases} \text{fuel_load} & \text{PAD data} \\ \min\left(1.0, \frac{\text{fuel_load}}{\text{max_fuel_value}}\right) & \text{legacy data} \end{cases} \quad (4)$$

PAD data derived from LiDAR are pre-normalized to the 0-1 range during preprocessing, representing the integrated plant area density where higher values indicate denser vegetation. The linear fuel relationship ($F_{\text{fuel}} = \text{fuel_load}$) was selected because PAD is already a physically meaningful measure of fuel availability—the integrated plant area per unit volume—requiring no additional transformation. The legacy fuel handling maintains backward compatibility with synthetic test cases and allows model testing without LiDAR data.

Wind Effect Factor Wind effects incorporate terrain-based amplification, particularly important for modeling fire behavior in Tenerife’s barrancos. The wind factor combines elevation and slope effects:

$$F_{\text{wind}} = F_{\text{elevation}} \times F_{\text{slope_wind}} \times F_{\text{aspect}} \quad (5)$$

where:

$$F_{\text{elevation}} = 1.0 + \frac{\text{elevation}}{1000.0} \times 0.1 \quad (6)$$

$$F_{\text{slope_wind}} = 1.0 + \text{slope} \times 0.5 \quad (7)$$

$$F_{\text{aspect}} = 1.0 \text{ (simplified)} \quad (8)$$

This formulation captures three key terrain-wind interactions observed in mountainous fire environments. The elevation effect ($F_{\text{elevation}} = 1.0 + \text{elevation}/1000 \times 0.1$) accounts for increased wind speeds at higher elevations due to reduced surface friction, producing 10% amplification per kilometer consistent with mountain terrain observations (Sharples, 2009). For Tenerife’s fire-affected elevations (0–2,200 m),

this yields factors from 1.0 to 1.22. The slope-wind interaction ($F_{\text{slope_wind}} = 1.0 + \text{slope} \times 0.5$) represents wind acceleration over ridges and through valleys, capturing the observed channeling where steep barrancos funnel winds and create localized fire acceleration zones. The aspect term is simplified to 1.0 because directional wind effects are captured through vector dot products in the full wind model. The composite wind factor is bounded to $[0.5, 2.0]$ to maintain realistic fire behavior—preventing unrealistic extremes while allowing upwind spread at reduced rates and downwind acceleration. Alternative formulations using computational fluid dynamics were rejected due to computational cost and lack of high-resolution wind field data for calibration.

Slope Effect Factor Slope effects incorporate topographic influences across Tenerife’s diverse elevation gradients using a trigonometric approach:

$$F_{\text{slope}} = 1.0 + \text{slope_influence} \times \sin(\theta_{\text{slope}}) \quad (9)$$

where θ_{slope} is calculated as:

$$\theta_{\text{slope}} = \arctan\left(\frac{\Delta\text{elevation}}{\text{horizontal_distance}}\right) \quad (10)$$

Distance correction accounts for increased path length in diagonal connections:

$$F_{\text{distance}} = \begin{cases} \frac{1}{\sqrt{2}} \approx 0.707 & \text{if diagonal spread} \\ 1.0 & \text{if orthogonal spread} \end{cases} \quad (11)$$

The sine function $\sin(\theta_{\text{slope}})$ represents the vertical component of slope angle, directly related to flame tilt and radiative heating of upslope fuels—the primary mechanism by which slope accelerates fire spread (as established by Rothermel, 1972; Sullivan, 2009). For upslope spread, positive $\Delta\text{elevation}$ increases F_{slope} above 1.0, while downslope spread reduces it below 1.0. For moderate slopes (30°), $\sin(30^\circ) = 0.5$ produces 10–25% spread modification (with $\text{slope_influence} = 0.2$ – 0.5), matching field observations of upslope

fire acceleration in Mediterranean and other forest types (Viegas, 2004). Alternative formulations using tangent ($\tan(\theta)$)—employed in Rothermel’s original model—were rejected because $\sin(\theta)$ remains bounded for steep slopes ($\sin(90^\circ) = 1.0$ vs. $\tan(90^\circ) \rightarrow \infty$), preventing numerical instabilities on Tenerife’s volcanic terrain where slopes frequently exceed 40° . The `slope_influence` coefficient is calibrated to account for fuel moisture, wind conditions, and vegetation structure specific to the 2023 Tenerife fire conditions. The calculated probability is bounded to $[0, 1]$ and compared against a random value for ignition determination, providing stochastic fire behavior.

3.5.2 Vertical Fire Spread

Vertical fire propagation enables fire to spread between adjacent forest layers, representing the critical transition from surface fire to crown fire. The model implements vertical spread through vertical connectivity—a pre-computed measure of fuel continuity between adjacent vertical layers derived from LiDAR-based Plant Area Density (PAD) data (Andersen et al., 2005; Riaño et al., 2003).

Calculating Vertical Connectivity Vertical fire spread in forests occurs when flames from a burning layer preheat and ignite fuels in adjacent layers above or below. Unlike horizontal spread, which is strongly influenced by wind and slope, vertical spread is primarily controlled by the physical continuity of fuel between layers—the presence of ladder fuels that bridge vertical gaps (Andersen et al., 2005; Riaño et al., 2003). Traditional fire models represent this through discrete fuel ladders or canopy base height thresholds, but these approaches cannot utilize the continuous vertical fuel profiles provided by LiDAR PAD data (Scott & Reinhardt, 2001). The connectivity-based approach directly translates PAD profiles into spread probabilities, ensuring that fire spreads vertically only where sufficient fuel exists in both source and target layers. This is particularly important for Tenerife’s Canary Island pine forests, where under-

story regeneration creates variable vertical fuel continuity—dense in some locations (facilitating crown fire) and sparse in others (limiting fire to surface layers). Alternative approaches considered included using fixed vertical spread probabilities or PAD thresholds, but these do not capture the continuous gradient of vertical fuel connectivity observed in heterogeneous forests. The connectivity formulation also ensures symmetry: upward and downward spread between the same two layers have identical base probabilities, consistent with the physical principle that fuel continuity is bidirectional.

Before simulation begins, vertical connectivity is pre-computed for every interface between adjacent layers across the entire simulation domain, eliminating redundant calculations during runtime and contributing to computational efficiency. For PAD data (already normalized to 0-1 range during preprocessing), fuel loads are used directly:

$$\text{connectivity}(x, y, z) = \sqrt{\text{fuel_lower}(x, y, z) \times \text{fuel_upper}(x, y, z + 1)} \quad (12)$$

The square root of the product (geometric mean) ensures that high connectivity requires substantial fuel in both adjacent layers, consistent with the physical requirement that fire must have fuel to burn in both the source layer (to generate sufficient heat) and target layer (to sustain ignition). This formulation has several desirable properties: (1) Connectivity equals zero if either layer has no fuel ($\sqrt{1.0 \times 0.0} = 0$), preventing fire spread across fuel gaps—critical for representing fuel discontinuities that arrest crown fire development. (2) Connectivity equals 1.0 only when both layers have maximum fuel ($\sqrt{1.0 \times 1.0} = 1.0$), representing optimal conditions for vertical fire propagation. (3) Connectivity for moderate fuel in both layers ($\sqrt{0.5 \times 0.5} = 0.5$) falls between these extremes, capturing gradual variation in spread potential. Alternative formulations were considered and rejected: the arithmetic mean $[(\text{fuel_lower} + \text{fuel_upper})/2]$ produces non-zero connectivity even when one

layer is empty (e.g., $[1.0 + 0.0]/2 = 0.5$), violating physical constraints; the minimum function $[\min(\text{fuel_lower}, \text{fuel_upper})]$ underestimates connectivity when fuels are unequal (e.g., $\min(0.8, 0.6) = 0.6$ vs. $\sqrt{0.8 \times 0.6} \approx 0.69$). The geometric mean formulation has precedent in canopy fuel connectivity studies and ladder fuel assessments using LiDAR data, though those studies used it for static fuel characterization rather than dynamic fire spread modeling (Kramer et al., 2014).

Vertical Ignition Probability When fire attempts to spread vertically from a burning cell to an unburned adjacent layer, the probability of ignition is determined directly from the pre-computed connectivity:

$$P_{\text{vertical}} = \text{vertical_connectivity}(x, y, z_{\text{interface}}) \quad (13)$$

where $z_{\text{interface}} = \min(z_{\text{source}}, z_{\text{target}})$ ensures consistent interface indexing for both upward and downward spread. The connectivity array stores 19 values for a model with 20 layers—one value for each interface between adjacent layers. $\text{Connectivity}[10]$ represents how well fuels connect between layer 10 and layer 11. Whether fire spreads upward (10→11) or downward (11→10), the model uses $\min(10, 11) = 10$ to look up the same connectivity value, ensuring that the fuel bridge between two layers works the same way regardless of spread direction—a physical requirement since the fuel structure itself doesn’t change based on spread direction.

Unlike horizontal spread, which requires multiplicative combination of base probability, fuel, wind, slope, and distance factors (Equation 3), vertical spread uses only the connectivity value as the ignition probability. This simplification is justified by the physics of vertical fire spread: wind effects are primarily horizontal, acting to tilt flames laterally rather than enhance vertical heat transfer between layers. Slope effects similarly influence horizontal spread direction but do not significantly alter the vertical fuel continuity that controls layer-to-layer fire propagation. By setting wind

and slope factors to 1.0 for vertical spread, the model correctly represents that vertical connectivity—determined by vertical fuel structure—is the dominant control on crown fire development (Cruz et al., 2004). The fuel factor is already incorporated within the connectivity calculation (as the geometric mean of adjacent layer fuels) rather than applied as a separate multiplier, avoiding double-counting of fuel effects. Alternative approaches that apply full horizontal spread factors (wind, slope) to vertical spread were tested during model development but produced unrealistic crown fire behavior, with excessive vertical spread in high-wind conditions even when vertical fuel continuity was low.

The calculated probability is compared against a random number between 0 and 1 for ignition determination. Using probability rather than fixed thresholds creates more realistic fire patterns—in real fires, crown fire doesn’t develop uniformly but spreads upward in some locations while staying on the ground in others, even where fuel conditions are similar. This variability occurs due to turbulent flames, local wind fluctuations, and small-scale fuel variations the model cannot resolve at 2 m vertical resolution. A deterministic approach would produce unrealistic results where fire either stays entirely on the surface or transitions to crown fire everywhere simultaneously. The probabilistic approach allows crown fire to develop gradually and unevenly across the landscape, matching the patchy crown fire patterns observed in the 2023 Tenerife fire where dense pine stands experienced crown fire while areas with fuel gaps remained as surface fire.

3.5.3 Ember Transport Mechanism

Ember transport enables long-distance fire spread beyond the Moore neighborhood, representing the critical spotting mechanism where embers ignite new fires hundreds of meters ahead of the main fire front. The model implements ember transport through a four-stage process: generation, travel, height dynamics, and ignition assessment.

Stage 1: Ember Generation Each burning cell has a probability of generating embers at each simulation time step:

$$P_{\text{ember gen}} = P_{\text{ember probability}} \times F_{\text{height}} \times F_{\text{wind gen}} \quad (14)$$

where the height factor accounts for increased ember generation at higher forest layers:

$$F_{\text{height}} = 1.0 + \frac{z}{\text{num_layers}} \times \text{ember_height_factor} \quad (15)$$

and the wind generation factor represents increased ember production under stronger winds:

$$F_{\text{wind gen}} = 1.0 + \frac{v_{\text{wind}}}{10.0} \times 0.1 \quad (16)$$

Upon successful generation, the number of embers produced follows a discrete uniform distribution:

$$N_{\text{embers}} \sim \text{DiscreteUniform}(1, 3) \quad (17)$$

The ember generation formulation captures three key physical processes controlling spotting in crown fires. First, the base ember probability $P_{\text{ember probability}}$ (a calibrated parameter, typical range 0.1–0.5) represents the intrinsic likelihood that a burning cell produces firebrands—influenced by fuel type, fire intensity, and combustion dynamics (Koo et al., 2010). Second, the height factor F_{height} increases generation probability linearly with vertical position, consistent with observations that crown fires produce more embers than surface fires due to higher plume buoyancy and larger fuel particles (bark, branches) becoming airborne (Albini & Intermountain Forest and Range Experiment Station (Ogden, 1979). For a 20-layer model, cells at the canopy top ($z = 19$) have $F_{\text{height}} = 1.0 + 19/20 \times \text{ember_height_factor}$, while surface cells ($z = 0$) have $F_{\text{height}} = 1.0$. Third, the wind generation factor $F_{\text{wind gen}}$

represents increased ember production under stronger winds, where higher wind speeds enhance fuel fragmentation and increase the number of particles lofted into the plume. The 10% increase per 10 m/s wind speed is conservative, preventing excessive ember generation while capturing the observed trend. The discrete uniform distribution for ember count (1–3 embers per successful generation) introduces stochastic variability in spotting patterns while maintaining computational tractability.

Stage 2: Ember Travel Distance and Direction Each ember travels a distance sampled from an exponential distribution:

$$d_{\text{ember}} = \max(1, \text{Exponential}(\lambda = \text{ember_distance})) \quad (18)$$

The travel direction incorporates wind bias through weighted interpolation between random and wind-aligned directions:

$$\theta_{\text{ember}} = (1 - w_{\text{strength}}) \times \theta_{\text{random}} + w_{\text{strength}} \times \theta_{\text{wind}} \quad (19)$$

where w_{strength} is the wind strength factor (range 0–1), θ_{random} is a uniform random angle, and θ_{wind} is the wind direction.

The exponential distance distribution captures the observed pattern that most embers land close to the source (short-range spotting) while rare embers travel long distances (long-range spotting), consistent with firebrand transport models and empirical spotting distance data (Albini & Intermountain Forest and Range Experiment Station (Ogden, 1979). The rate parameter $\lambda = \text{ember_distance}$ (typical value 5 cells = 100 m at 20 m resolution) controls the mean spotting distance, with smaller values producing longer-range spotting. The minimum distance of 1 cell prevents embers from landing at the source cell. Alternative distance distributions were considered: uniform distributions produce unrealistic equal probabilities at all distances; normal distributions underrepresent long-distance spotting; power-

law distributions better match extreme spotting events but require careful parameterization to avoid numerical issues. The exponential distribution provides a simple, physically reasonable compromise validated in multiple firebrand transport studies (Koo et al., 2010). The directional interpolation between random (θ_{random}) and wind-aligned (θ_{wind}) angles captures the stochastic nature of ember transport, where turbulent eddies cause some embers to travel perpendicular or even upwind, while the bulk drift downwind. The wind strength weight w_{strength} (derived from wind speed, typically 0.4–0.8) controls the degree of wind bias.

Stage 3: Ember Height Dynamics Embers can move vertically between forest layers during transport:

$$\Delta z = \begin{cases} \text{Uniform}(-2, \text{ember_rise}) & \text{if ember_rise} \\ & \text{configured} \\ \text{Uniform}(-1, 1) & \text{otherwise} \end{cases} \quad (20)$$

The target layer is bounded to valid indices:

$$z_{\text{target}} = \max(0, \min(\text{num_layers} - 1, z_{\text{source}} + \Delta z)) \quad (21)$$

Vertical ember dynamics represent the trajectory of firebrands through the forest canopy—embers generated at the canopy top may rise further due to plume buoyancy (positive Δz) or descend toward the surface (negative Δz) as they cool and lose lift. The configurable ember rise parameter (typical value 2 layers = 4 m) allows embers to ascend above their generation height, consistent with observations that strong fire plumes loft embers 10–50 m above the canopy. Downward motion (up to -2 layers = -4 m) represents descending embers as they lose buoyancy and aerodynamic lift. The uniform distribution is a simplification—realistic ember trajectories follow complex parabolic paths determined by plume updrafts, particle size, and atmospheric conditions (Sardoy et al., 2007)—but captures

the essential feature that embers can land at various vertical positions, enabling ignition of surface fuels (low layers) or direct canopy ignition (high layers). The bounding operation ensures embers cannot land below ground ($z = 0$) or above the canopy ($z = \text{num_layers} - 1$).

Stage 4: Ember Ignition Assessment

The probability that a landing ember successfully ignites the target cell is:

$$P_{\text{ignition}} = P_{\text{ember ignition}} \times F_{\text{fuel}} \times F_{\text{moisture}} \times F_{\text{height}} \times F_{\text{distance}} \quad (22)$$

where:

$$F_{\text{fuel}} = \min\left(1.0, \frac{\text{fuel_load}}{\text{max_fuel_value}}\right) \quad (23)$$

$$F_{\text{moisture}} = 1.0 - \text{fuel_moisture} \quad (24)$$

$$F_{\text{height}} = \begin{cases} 1.2 & \text{if } z < \text{num_layers}/2 \\ & \text{(lower layers)} \\ 0.8 & \text{otherwise (upper layers)} \end{cases} \quad (25)$$

$$F_{\text{distance}} = \max(0.3, 1.0 - \frac{d_{\text{ember}}}{2 \times \text{ember_distance}}) \quad (26)$$

$$\frac{d_{\text{ember}}}{2 \times \text{ember_distance}} \quad (27)$$

The ember ignition probability formulation balances physical realism with computational simplicity. The base ember ignition probability $P_{\text{ember ignition}}$ (calibrated parameter, typical range 0.2–0.4) represents the intrinsic likelihood that a landing ember transitions from smoldering to flaming ignition, influenced by ember temperature, size, and fuel bed receptivity (Koo et al., 2010). The fuel factor F_{fuel} ensures embers ignite only where sufficient fuel exists—bare ground or fuel gaps prevent spotting success. The moisture factor F_{moisture} reduces ignition probability in moist fuels, consistent with observations that spotting success depends critically on fuel moisture content. The height factor F_{height} represents different ignition success rates by layer: lower layers (surface fuels) have 20% higher ignition success ($F_{\text{height}} = 1.2$) due to finer fuels and accumulated litter, while upper layers (canopy) have

20% lower success ($F_{\text{height}} = 0.8$) due to coarser fuels and moisture retention. The distance decay factor F_{distance} represents ember cooling during flight—embers traveling farther lose temperature and become less likely to ignite upon landing. The linear decay to 0.3 at twice the mean spotting distance prevents complete failure at long ranges (consistent with rare long-distance spotting events) while preferentially favoring near-source ignitions. For the 2023 Tenerife fire, this formulation produced realistic spotting patterns with most spot fires occurring 50–200 m downwind, matching EMSR-observed fire perimeter geometry.

3.5.4 Terrain-Fire Interactions

The model incorporates terrain effects on fire spread through three mechanisms: directional slope effects, general terrain wind modifications, and barranco-specific wind channeling.

Directional Slope Effects on Fire Spread

For each potential fire spread between cells, the slope angle in the direction of spread is calculated using the arctangent of elevation change over horizontal distance:

$$\theta_s = \arctan 2(\Delta h, d_h) \quad (28)$$

where Δh is the elevation difference between target and source cells (m), and d_h is the horizontal distance (20 m for orthogonal, $20\sqrt{2}$ m for diagonal neighbors).

The slope factor modifying spread probability is:

$$F_s = 1.0 + s_i \times \sin(\theta_s) \quad (29)$$

where s_i is the slope influence coefficient (calibrated parameter, typical range 0.2–0.5). The slope factor is bounded to prevent unrealistic downslope suppression:

$$F_{s,\text{final}} = \max(0.1, F_s) \quad (30)$$

The directional slope formulation calculates the actual terrain slope between source and target cells rather than using pre-computed slope grids, ensuring accuracy for diagonal spread where the slope direction may not align

with cardinal directions. The arctangent function $\arctan 2(\Delta h, d_h)$ computes the slope angle θ_s (in radians), which is positive for upslope spread and negative for downslope spread. Using $\sin(\theta_s)$ in the slope factor provides linear slope effects for moderate angles while remaining bounded for steep slopes ($\sin(90^\circ) = 1.0$), preventing numerical instabilities on volcanic terrain where slopes frequently exceed 40° (Rothermel, 1972; Viegas, 2004). The slope influence coefficient s_i scales the magnitude of terrain effects and is determined through calibration. For moderate slopes (30°), $\sin(30^\circ) = 0.5$ produces 10–25% spread modification with $s_i = 0.2$ –0.5, matching field observations of upslope fire acceleration in Mediterranean forests (Viegas, 2004). The lower bound of 0.1 ensures fire can spread downslope at reduced rates (90% reduction from baseline) rather than being completely suppressed, consistent with observations that intense fires can spread downslope when fire intensity is high. Alternative formulations using $\tan(\theta_s)$ were rejected due to unbounded behavior at steep angles approaching vertical. The directional calculation is essential for accurately representing fire spread across complex barranco systems, where local slope direction varies significantly over short distances.

General Terrain Wind Modifications

Wind speed is modified by terrain through two elevation-dependent mechanisms representing atmospheric boundary layer effects and topographic acceleration.

Slope-based Wind Acceleration:

Wind speed increases over steep terrain due to flow compression and acceleration:

$$F_{\text{slope}} = 1.0 + 0.3 \times \text{clip}\left(\frac{\theta_{\text{slope}}}{60^\circ}, 0, 1\right) \quad (31)$$

where θ_{slope} is the local terrain slope angle and the clip function bounds the normalized slope to $[0, 1]$.

Elevation-based Wind Acceleration:

Wind speed increases with elevation due to reduced surface friction:

$$F_{\text{elev}} = 1.0 + 0.2 \times \frac{E - E_{\min}}{E_{\max} - E_{\min}} \quad (32)$$

where E is the cell elevation, $E_{\min}$ is the minimum domain elevation, and $E_{\max}$ is the maximum domain elevation.

The slope-based wind acceleration represents flow compression over ridges and steep terrain, where wind speeds can increase by 30% over the steepest slopes (60° or greater). The coefficient 0.3 indicates maximum 30% amplification, consistent with observations of topographic wind acceleration in mountainous terrain (Sharples, 2009). The normalization by 60° ensures that moderate slopes (30°) produce 15% amplification, while slopes exceeding 60° reach maximum amplification. The elevation-based acceleration represents the atmospheric boundary layer effect, where wind speeds increase with height above the surface due to reduced friction. The coefficient 0.2 produces 20% amplification from lowest to highest elevations in the domain, yielding gradual wind speed increase with elevation. The linear relationship is a simplification of logarithmic wind profiles but captures the essential trend while maintaining computational simplicity. Alternative formulations using logarithmic profiles were considered but require specification of surface roughness parameters not available from the terrain data. These general terrain modifications apply uniformly across the landscape and are distinct from the localized barranco channeling effects described below.

Barranco Wind Channeling Barrancos (volcanic ravines) modify wind behavior through two mechanisms that capture the distinctive channeling effects observed in steep-sided valleys.

Wind Speed Amplification in Barrancos:

Wind speed increases in barranco areas due to channeling through narrow valleys:

$$v_{\text{barranco}} = v_{\text{base}} \times f_{\text{amp}} \quad (33)$$

where v_{base} is the base wind speed and f_{amp}

is the amplification factor (typical value 2.0–3.0, representing 100–200% increase).

Wind Direction Alignment with Barranco Orientation:

Wind direction is progressively aligned with the barranco’s natural flow direction through weighted interpolation:

$$w_{\text{eff}} = f_{\text{barranco}} \times s_{\text{terrain}} \quad (34)$$

$$\theta_{\text{final}} = (1 - w_{\text{eff}}) \times \theta_{\text{wind}} + w_{\text{eff}} \times \theta_{\text{barranco}} \quad (35)$$

where w_{eff} is the effective barranco influence weight, f_{barranco} is the barranco influence parameter (0–1), s_{terrain} is the terrain effect strength (calibrated parameter), θ_{wind} is the meteorological wind direction, and θ_{barranco} is the barranco’s flow direction derived from terrain aspect.

The barranco wind channeling mechanisms represent the distinctive fire behavior observed in volcanic ravines, where steep valley walls funnel and accelerate winds, creating fire acceleration zones that contributed significantly to the 2023 Tenerife fire’s rapid spread. The amplification factor f_{amp} (typical range 2.0–3.0) represents observed wind speed increases in confined valleys, where valley geometry creates venturi effects that locally double or triple wind speeds compared to open terrain. Studies of wind flow in mountainous terrain document amplification factors of 2–4 in steep valleys, depending on valley width and wall steepness (Sharples, 2009). For barrancos typically 50–200 m wide with slopes exceeding 30°, amplification factors of 2.0–2.5 are physically reasonable. The directional alignment mechanism captures the observation that winds flowing through confined valleys are progressively turned to align with valley orientation, regardless of the synoptic wind direction. The effective weight w_{eff} controls the degree of alignment: values near 0 indicate weak channeling (wind retains original direction), while values near 1 indicate strong channeling (wind fully aligned with barranco). The product $f_{\text{barranco}} \times s_{\text{terrain}}$ combines the barranco-specific influence (varying spatially based on barranco geometry) with the overall terrain effect strength (a global

calibrated parameter). The barranco flow direction θ_{barranco} is derived from the local terrain aspect in barranco areas, representing the downslope drainage direction that winds naturally follow. These formulations are essential for modeling fire behavior where barrancos act as fire corridors. Alternative approaches using computational fluid dynamics to model barranco winds were considered but rejected due to computational cost and lack of high-resolution wind field validation data.

3.6 Model Calibration Framework

Calibration and Validation Target Creation The model framework incorporates 17 adjustable parameters governing fire behavior across core fire mechanics, environmental interactions, wind dynamics, and ember transport systems (Table 2). A two-stage calibration approach was implemented: (1) global sensitivity analysis to rank behavior-affecting model parameters by their influence on fire spread metrics, and (2) focused grid search calibration on the top 4-5 most influential parameters.

Existing operational models often have fixed parameters or limited calibration interfaces designed for standard fuel models developed for North American or Mediterranean ecosystems (Pavlidou, 2011; Rothermel, 1972). However, *Pc* fire behavior requires region-specific parameter estimation rather than generic literature values (Climent et al., 2004). The 2023 Tenerife fire (EMSR 685) demonstrated extreme fire behavior driven by low relative humidity, elevated temperatures, calima winds, and exceptionally dry fuel conditions following prolonged drought (Brunat et al., 2023).

Comprehensive grid search across all 17 model parameters at 3 points per parameter would require $3^{16} \approx 43$ million simulations. The multi-stage approach uses systematic one-at-a-time (OAT) sensitivity analysis to identify the 4-5 most influential parameters, reducing calibration to $3^4 = 81$ to $3^5 = 243$ simulations—a six-order-of-magnitude reduction while targeting parameters that control model behavior (hambyReviewTechniquesMeasurement1994).

This is essential for custom models where existing parameter sets cannot be transferred from other regions. The focused grid search explores 3-5 values per parameter spanning literature-supported ranges, balancing thorough coverage with computational feasibility while ensuring systematic exploration and reproducible results. (Sullivan, 2009)

3.6.1 Parameter Identification

The framework identifies 16 behavior-affecting parameters focusing on core fire mechanics (spread probability, fuel consumption rate, ignition threshold), environmental interactions (wind and slope influence coefficients), and ember transport systems (ember generation and ignition probabilities), while maintaining infrastructure settings (grid resolution, time step, random seed) constant. Parameters were selected based on three criteria: (1) direct appearance in fire spread probability calculations; (2) physical interpretability and calibratable nature (avoiding derived quantities like connectivity); and (3) literature precedent for calibration in fire behavior models (Pavlidou, 2011). Alternative approaches including all 16+ model parameters were rejected because they would include numerically derived values (e.g., maximum fuel value for normalization) or geometric parameters (number of layers, cell size) that are data-driven rather than behaviorally tunable. The 16-parameter set balances comprehensive coverage of fire behavior mechanisms with computational tractability.

3.6.2 Range-Based Sensitivity Analysis

The sensitivity analysis employs a standardized range-based approach evaluating each parameter at 9 evenly distributed positions across its full feasible range (0%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%) without requiring baseline dependency.

Implementation Specifications:

- 16 parameters $\times$ 9 evaluation points = 144 simulations

The range-based approach with 9 evenly spaced points provides systematic coverage of each parameter’s full feasible range, enabling

Step	Parameter Name	Range	Category
1	spread_probability	0.6-0.95	Core Fire Mechanics
2	fuel_consumption_rate	0.3-0.8	Core Fire Mechanics
3	ignition_threshold	0.01-0.5	Core Fire Mechanics
4	min_fuel_value	0.02-0.10	Core Fire Mechanics
5	max_fuel_value*	0.5-10.0	Core Fire Mechanics
6	wind_influence_on_spread	0.0-1.0	Environmental Interactions
7	slope_influence	0.0-1.0	Environmental Interactions
8	reference_wind_speed	5.0-25.0 m/s	Environmental Interactions
9	fuel_moisture_baseline	0.1-0.8	Environmental Interactions
10	wind_speed	1.0-20.0 m/s	Wind Parameters
11	wind_direction	0-360°	Wind Parameters
12	ember_probability	0.2-0.6	Ember Mechanics
13	ember_distance	2-15 cells	Ember Mechanics
14	ember_ignition	0.3-0.8	Ember Mechanics
15	ember_height_factor	0.0-1.0	Ember Mechanics
16	ember_wind_factor	0.0-1.0	Ember Mechanics
17	ember_rise	1-10 cells	Ember Mechanics

Figure 3: Model Calibration Parameters. The model framework incorporates 17 adjustable parameters across four categories: Core Fire Mechanics (5 parameters), Environmental Interactions (4 parameters), Wind Parameters (2 parameters), and Ember Mechanics (6 parameters). Of these, 16 are behavior-affecting parameters with `max_fuel_value` excluded. Sensitivity analysis evaluated 13 parameters, with the top 4-5 most sensitive parameters used for focused grid search calibration.

robust sensitivity estimation without baseline dependency. Alternative baseline-perturbation methods ($\pm 10\%$, $\pm 20\%$) were rejected due to baseline sensitivity and potential to miss nonlinear responses at parameter range extremes. The sampling provides sufficient resolution to detect nonlinear sensitivity patterns while maintaining computational feasibility (153 simulations vs. 1,000+ for finer resolution). Studies indicate that 5–10 sampling points per parameter provide adequate sensitivity estimation for most nonlinear systems (Hamby, 1994). The range-based method is particularly appropriate for the custom fire model where optimal parameter values are unknown a priori—sampling the full parameter

space ensures sensitivity rankings reflect global rather than local parameter importance. *Two-Stage Sensitivity Assessment:*

Stage 1: Fire Behavior Quantification

Each simulation is evaluated using a composite fire behavior objective capturing dynamic fire characteristics through four normalized components ($C_i \in [0, 1]$):

$$\text{Obj}_{\text{sens}} = \sum_i w_i C_i = 0.3C_{\text{area}} + 0.3C_{\text{rate}} + 0.2C_{\text{persist}} + 0.2C_{\text{dispersion}} \quad (36)$$

where weights w_i sum to unity and em-

phasize fire magnitude and spreading velocity equally (30% each), with secondary components for temporal persistence and spatial distribution (20% each).

Area Component (30% weight): Fire impact magnitude

$$C_{\text{area}} = \min \left(\frac{N_{\text{burned}}}{N_{\text{total}}}, 1.0 \right) \quad (37)$$

where N_{burned} is the total number of burned cells accumulated during simulation and N_{total} is the total grid size. This component quantifies cumulative fire extent relative to available domain.

Rate Component (30% weight): Fire spreading intensity

$$C_{\text{rate}} = \min \left(\frac{N_{\text{burned}}}{t_{\text{sim}} \cdot N_{\text{total}}}, 1.0 \right) \quad (38)$$

where t_{sim} is the simulation duration in time steps. This component quantifies fire propagation rate (burned cells per time step) normalized by domain size.

Persistence Component (20% weight): Simulation duration relative to maximum allowed

$$C_{\text{persist}} = \min \left(\frac{t_{\text{sim}}}{t_{\text{max}}}, 1.0 \right) \quad (39)$$

where t_{max} is the maximum allowed simulation time steps. This component quantifies temporal fire activity duration relative to simulation limits.

Dispersion Component (20% weight): Spatial fire distribution geometry

$$C_{\text{dispersion}} = \min \left(\frac{1}{2} \left(\frac{\sigma_x}{W} + \frac{\sigma_y}{H} \right), 1.0 \right) \quad (40)$$

where σ_x and σ_y are the standard deviations of burned cell coordinates in x and y directions, and W and H are the grid dimensions (width and height in cells). This component quantifies spatial fire spread geometry using coefficient of variation principles, producing values near 0 for concentrated fires and approaching 1.0 for fires distributed across the entire domain.

The sensitivity analysis objective intentionally differs from calibration (spatial error

against EMSR data), aiming to identify parameters producing variability in fire behavior rather than matching a specific target. The composite objective weights burned area and spread rate (30% each) for fire magnitude and intensity, with persistence and dispersion (20% each) capturing temporal and spatial characteristics. This ensures sensitivity rankings reflect parameters that meaningfully alter fire behavior across multiple dimensions rather than optimizing for a single metric that might produce degenerate rankings. The equal weighting of area and rate emphasizes that parameters affecting both fire extent and speed are most critical. Alternative single-metric approaches (e.g., burned area only) were tested but produced unreliable rankings where parameters affecting spread rate but not final area were incorrectly identified as insensitive.

Stage 2: Parameter Sensitivity Ranking

For each parameter i , the 9 objective values obtained from range-based sampling are processed through the sensitivity index formula:

$$S_i = \frac{\text{Range}(\text{Obj}_i)}{\text{Range}(P_i)} \times \frac{1}{|\text{Median}(\text{Obj}_i)|} \quad (41)$$

where:

- S_i is the dimensionless sensitivity index for parameter i
- $\text{Range}(\text{Obj}_i) = \max(\text{Obj}_i) - \min(\text{Obj}_i)$ is the range of objective values across 9 test points
- $\text{Range}(P_i) = P_{i,\text{max}} - P_{i,\text{min}}$ is the parameter range explored
- $\text{Median}(\text{Obj}_i)$ is the median objective value for parameter i

The sensitivity index S_i provides dimensionless, cross-parameter ranking by normalizing output variability by input variability and median output magnitude. This formulation addresses three methodological challenges: (1) **Scale normalization**—parameters with different units and scales (probabilities [0,1] vs. distances in meters) are made comparable through range normalization; (2) **Magnitude normalization**—division by median output prevents parameters with small absolute

effects but high relative effects from dominating rankings; (3) **Outlier robustness**—range-based metrics are more robust to outliers compared to variance-based approaches.

Parameters with high S_i values produce large changes in fire behavior relative to input variation, indicating strong control over model outputs and prioritization for calibration. The approach successfully identified `spread_probability`, `fuel_consumption_rate`, `ember_probability`, and `ember_ignition` as the top 4 most sensitive parameters (sensitivity indices $S = 1.4, 0.16, 0.12, 0.06$ respectively), with `spread_probability` demonstrating 13× higher sensitivity than the second-ranked parameter.

3.6.3 Focused Grid Search

Based on sensitivity results, calibration explored the top 4 most sensitive parameters: `spread_probability`, `fuel_consumption_rate`, `ember_probability`, and `ember_ignition`. Additionally, `min_fuel_value` was included based on its critical role in ignition threshold determination.

Grid Search Implementation:

- **Parameter Ranges:** Literature-informed bounds:
 - $P_{\min_fuel} \in [0.02, 0.10]$ (probability)
 - $P_{\text{spread}} \in [0.6, 0.95]$ (probability)
 - $P_{\text{fuel_rate}} \in [0.3, 0.8]$ (units/step)
 - $P_{\text{ember}} \in [0.2, 0.6]$ (probability)
- **Grid Resolution:** $n_{\text{points}} = 3$ points per parameter dimension
- **Total Combinations:** $n_{\text{total}} = 3^4 = 81$ parameter sets
- **Execution:** Parallel processing with 60 workers
- **Grid Size:** 609×609 cells (20 m resolution)

Limiting calibration to 4 parameters provides optimal balance between parameter space coverage and computational feasibility. With 3 points per parameter, 4 parameters yield 81 combinations, while 5 parameters would yield 243 combinations—a 3× increase for marginal

improvement given that the 5th parameter (`fuel_moisture_baseline`, $S = 0.04$) has 15× lower sensitivity than the top-ranked parameter. The 3-point grid resolution samples parameter space at minimum, midpoint, and maximum values, ensuring coverage of range extremes while maintaining tractable combinations.

Spatial Error Objective:

The calibration objective function combines spatial similarity and over-prediction penalties:

$$D = \frac{2|A \cap B|}{|A| + |B|} \quad (\text{Dice coefficient}) \quad (42)$$

$$E_{\text{spatial}} = 1 - D \quad (\text{Spatial error}) \quad (43)$$

$$E_{\text{penalty}} = \begin{cases} 2.0 \cdot (r - 2.0)^2 & \text{if } r > 2.0 \\ 0 & \text{otherwise} \end{cases} \quad (44)$$

$$\text{Obj}_{\text{calib}} = E_{\text{spatial}} + E_{\text{penalty}} \quad (45)$$

where:

- A represents the set of predicted burned cells from simulation
- B represents the set of observed burned cells from EMSR data
- $|A \cap B|$ is the intersection (correctly predicted burned cells)
- $|A|$ and $|B|$ are the cardinalities (total burned cells) of each set
- $D \in [0, 1]$ is the Dice coefficient measuring spatial overlap
- r is the area ratio quantifying over-prediction magnitude
- $\text{Obj}_{\text{calib}}$ is the total calibration objective (lower values indicate better performance)

Spatial Error Component (E_{spatial}): Quantifies spatial mismatch by converting the Dice similarity coefficient to an error metric. Values range from 0 (perfect spatial match) to 1 (no spatial overlap).

Over-prediction Penalty (E_{penalty}): Applies quadratic penalty for simulations exceeding twice the observed burned area. This component prevents convergence to unrealistic high-spread parameter sets that achieve spatial coverage through excessive burning.

The parameter combination achieving lowest $\text{Obj}_{\text{calib}}$ value is selected as optimal, ensuring spatial error minimization rather than

similarity maximization. The calibration objective minimizes spatial error (1 - Dice coefficient) rather than maximizes similarity, addressing a critical methodological issue identified during framework development. Preliminary calibration using similarity maximization produced counterintuitive results where parameters causing extreme over-prediction (burned area 2–3× observed) achieved high similarity scores due to complete target area coverage plus extensive additional burning.

The error minimization formulation correctly penalizes both under-prediction and over-prediction through the Dice coefficient’s symmetrical treatment of false positives and false negatives. Dice coefficient was selected as the sole similarity metric due to its smoother optimization landscape—test calibrations using multi-metric objectives showed tendency to converge to extreme parameter values achieving local optima in one metric while performing poorly in others. Dice coefficient alone provides robust, single-objective optimization suitable for grid search

The over-prediction penalty threshold of $r = 2.0$ was selected based on validation testing where area ratios < 2.0 indicated reasonable model behavior even with spatial misalignment, while ratios > 2.0 indicated fundamental over-prediction requiring strong correction. The combined objective successfully identified parameter sets producing spatial errors of 0.40–0.60 (Dice coefficients 0.40–0.60) without extreme over-prediction, matching observed fire perimeter geometry while maintaining physical realism.

3.7 Model Validation Framework

The validation process employs temporal split-sample validation using identical simulation configurations and evaluation metrics as calibration, ensuring performance assessment reflects true predictive capability rather than methodological artifacts.

Parameters calibrated on early fire progression (Days 1–2: August 18–21, 2023) are validated against later development (Days 3–4: August 24–26, 2023). This temporal separation

tests parameter generalization across different fire stages, weather conditions, and fuel consumption states. The 9-day observation period captures critical fire behavior transitions from initial spread through mature fire development.

Validation Protocol Simulations execute using optimized parameters from grid search calibration with identical preprocessing (LiDAR vegetation data, terrain features) and simulation engine configuration. Grid size is dynamically determined from fire perimeter extent with 20m resolution and maximum 100 timesteps for computational efficiency.

Performance Evaluation Validation employs three complementary assessment approaches:

Spatial Accuracy: Uses the identical spatial error objective as calibration **Fire Behavior Analysis:** Mechanistic validation through spread dynamics:

$$P_{\text{vertical}} = \frac{N_{\text{vertical}}}{N_{\text{total}}} \times 100\% \quad (46)$$

$$P_{\text{ember}} = \frac{N_{\text{ember}}}{N_{\text{total}}} \times 100\% \quad (47)$$

$$\eta_{\text{spread}} = \frac{N_{\text{successful}}}{N_{\text{attempted}}} \times 100\% \quad (48)$$

where N_{vertical} and N_{ember} represent vertical and ember-driven ignition events respectively, N_{total} is total ignitions, $N_{\text{successful}}$ is successful ignitions, and $N_{\text{attempted}}$ is attempted ignitions.

Vertical spread percentage (P_{vertical}) quantifies ground-to-crown fire transitions, indicating crown fire activity critical for forest fire behavior. Ember contribution percentage (P_{ember}) measures spotting-driven ignitions that enable rapid expansion across fuel breaks. Spread efficiency (η_{spread}) validates parameter plausibility by measuring ignition success rates—ensuring fire spreads under favorable conditions but is suppressed by terrain barriers and fuel gaps.

Expected ranges for Canary Pine forests include $P_{\text{vertical}} \in [30, 50]\%$ and $P_{\text{ember}} \in [10, 20]\%$. These metrics complement spatial accuracy and prevent equifinality where multiple parameter sets achieve similar spatial

performance through different physical mechanisms.

Computational Performance: Execution time tracking with memory optimization validation and sparse storage efficiency assessment.

Data Processing EMSR shapefiles are rasterized to match the simulation grid resolution using UTM Zone 28N projection, eliminating validation errors from geometric transformations. The multi-day dataset enables parameter consistency assessment across evolving fire conditions, critical for cellular automata models where parameters must remain physically meaningful across different spread rates.

4 Results

The evaluation of the LiDAR-integrated fire simulation model encompasses sensitivity analysis, parameter calibration, and temporal validation. Results demonstrate a hierarchical parameter sensitivity structure, successful calibration convergence, and critical overfitting challenges that reveal fundamental limitations in purely numerical optimization approaches.

4.1 Sensitivity Analysis Results

4.1.1 Parameter Sensitivity Hierarchy

The sensitivity analysis revealed extreme parameter dominance within the model structure. Five parameters demonstrated significant influence on fire behavior, with **spread_probability** showing $8.7\times$ greater influence than the second-ranked parameter (Table 2, Fig. 4).

Table 2: Parameter Sensitivity Ranking

Rank	Parameter	Sensitivity Index
1	spread_probab	1.406
2	fuel_consumption_rat	0.161
3	ember_probab	0.116
4	ember_ignition	0.056
5	fuel_moisture	0.038

4.1.2 Mechanistic Interpretation

The sensitivity hierarchy reveals three key insights:

Spread Probability Dominance: The exceptional sensitivity of **spread_probability** (1.4063) indicates fundamental cell-to-cell propagation rules control fire behavior more than environmental or fuel-specific factors.

Ember System Significance: Combined ember-related parameters (**ember_probability**: 0.1159, **ember_ignition**: 0.0564) totaling 0.1723 establish long-range transport as the second most important behavioral component.

Fuel System Influence: Parameters governing fuel consumption (0.1609) and moisture content (0.0382) confirm importance of accurate fuel characterization, though substantially less influential than spread and ember mechanisms.

Remaining parameters show negligible sensitivity (<0.015), supporting focused calibration strategies targeting the top 4-5 most influential parameters.

4.2 Calibration Results

Following sensitivity analysis, calibration targeted the top 4 most sensitive parameters using a 3-point grid search (81 combinations). The process identified two competing configurations with critical behavioral differences (Table 3).

The significant difference in **spread_probability** (0.95 vs. 0.775) revealed fundamental methodology limitations. While Rank 1 achieved lowest calibration error, preliminary testing demonstrated unrealistically rapid fire spread exceeding observed behavior patterns (Fig. 5). This necessitated selection of Rank 2 parameters despite higher numerical error—prioritizing physical realism over optimization metrics.

4.3 Validation Results

Final validation employed Rank 2 parameters against EMSR data using temporal split protocol (Days 1-2 calibration, Days 3-4 validation). Each simulation used $609 \times 609 \times 20$ layer grids

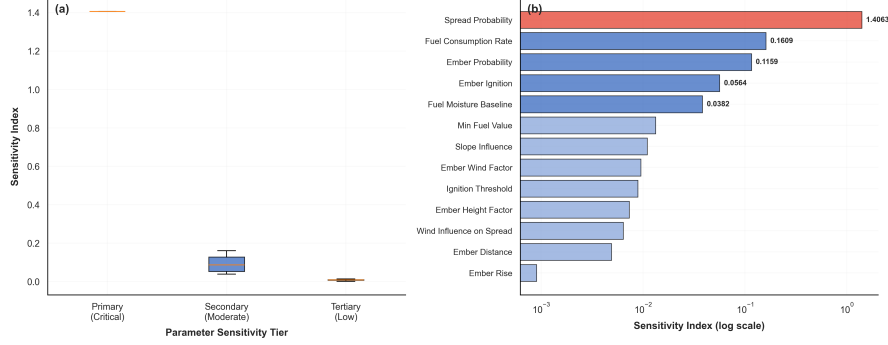


Figure 4: Sensitivity distribution analysis. (a) Box plots of parameter sensitivity indices grouped by tier (b) Per-parameter sensitivity indices on a log scale.

Table 3: Calibration Parameter Comparison. Rank 2 selected despite higher error due to realistic fire behavior.

Parameter	Rank 1 (Error: 0.513)	Rank 2 (Error: 0.536)	Rationale
min_fuel_value	0.02	0.02	Identical aggressive threshold
spread_probability	0.95	0.775	Rank 2 more realistic
fuel_consumption_rate	0.30	0.30	Identical slow consumption
ember_probability	0.60	0.60	Identical high generation

at 20m resolution, representing approximately 148 km² of fire-affected terrain.

4.3.1 Validation Analysis

Comprehensive validation results (Table 4) reveal critical model limitations through mechanistic metrics framework. Bolded values indicate substantial deviations from expected ranges.

These metrics reveal critical limitations: vertical spread (3.8%) falls far below Canary Pine forest ranges (30-50%), while ember contributions (43%) substantially exceed realistic spotting behavior (10-20%). Spread efficiencies >100% indicate uncontrolled propagation inconsistent with natural barriers. The consistency across both validation days indicates systematic model behavior (Fig. 6).

Despite 29.6% objective function improvement (Day 3 to Day 4), mechanistic validation revealed systematic overfitting:

- Ember efficiency >150% indicates uncon-

trolled long-range spotting

- Excessive ember contribution (43% vs. expected 10-20%)
- Insufficient crown fire activity (3.8% vs. expected 30-50%)
- Parameter optimization converged on compensating errors rather than physically accurate mechanisms

Visual examination (Fig. 7) confirms systematic overfitting through omnidirectional expansion beyond EMSR perimeters and unrealistic circular patterns ignoring topographic constraints.

4.3.2 Validation Framework Assessment

Results demonstrate both the effectiveness of temporal split-sample validation and the critical importance of mechanistic metrics. While achieving moderate spatial accuracy (0.49-0.69), the model exhibits systematic overfitting through unrealistic spread mechanisms. The

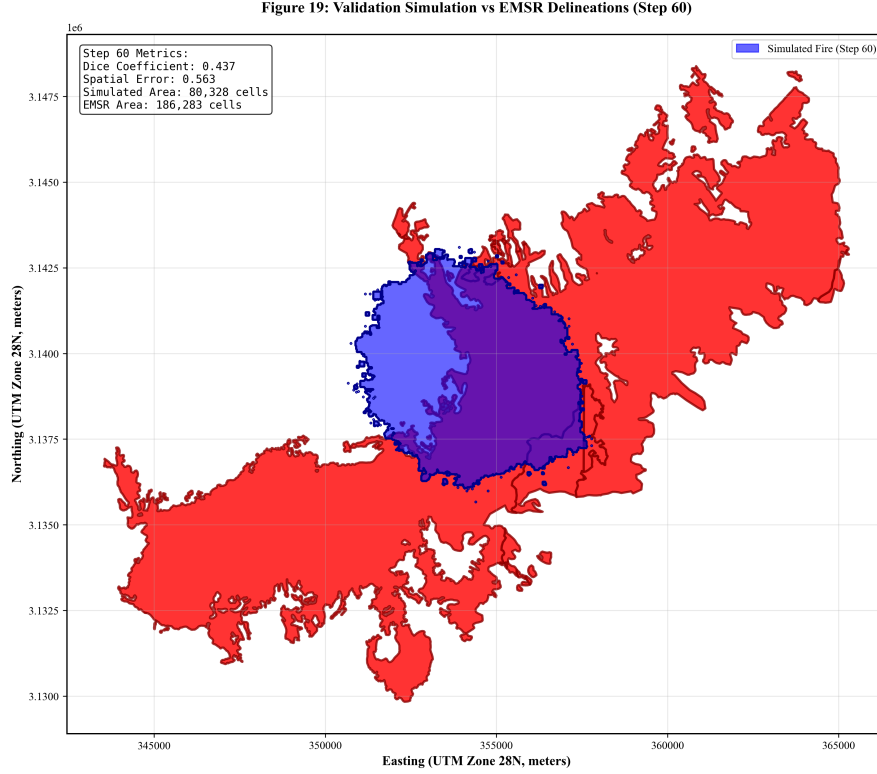


Figure 5: Rank 1 parameter validation showing unrealistic expansion beyond observed Day 4 EMSR fire perimeter

dominance of ember transport (43% vs. 10-20%) combined with insufficient crown fire activity (3.8% vs. 30-50%) validates the necessity of multi-metric validation frameworks that prevent equifinality and ensure predictions generalize through correct physical mechanisms rather than numerical artifacts.

Computational performance remained excellent (average 53.7 minutes per scenario), demonstrating practical applicability despite behavioral limitations.

5 Discussion

5.1 Addressing Research Question 1: LiDAR Integration for Enhanced Fuel Structure Representation

Successful LiDAR-to-CA Integration Framework The first research question asked how high-resolution LiDAR data can be used as input for CA wildfire models to represent vertical fuel structure. This study demonstrated that LiDAR data can be effectively integrated into CA frameworks through systematic pre-processing that converts 3D point clouds into spatially explicit fuel metrics, establishing a robust foundation for enhanced fire behavior modeling (Andersen et al., 2005; Coen et al., 2018).

The Beer-Lambert law implementation

Table 4: Comprehensive Validation Results and Fire Behavior Analysis

Validation Metric	Day 3	Day 4	Expected Range
<i>Spatial Performance</i>			
Objective Value	0.690	0.486	–
Execution Time (min)	53.2	54.1	–
Improvement	–	29.6%	–
<i>Fire Behavior Metrics</i>			
P_{vertical} (%)	3.84	3.82	30–50
P_{ember} (%)	43.0	43.0	10–20
η_{spread} (ember, %)	150.7	150.7	<100

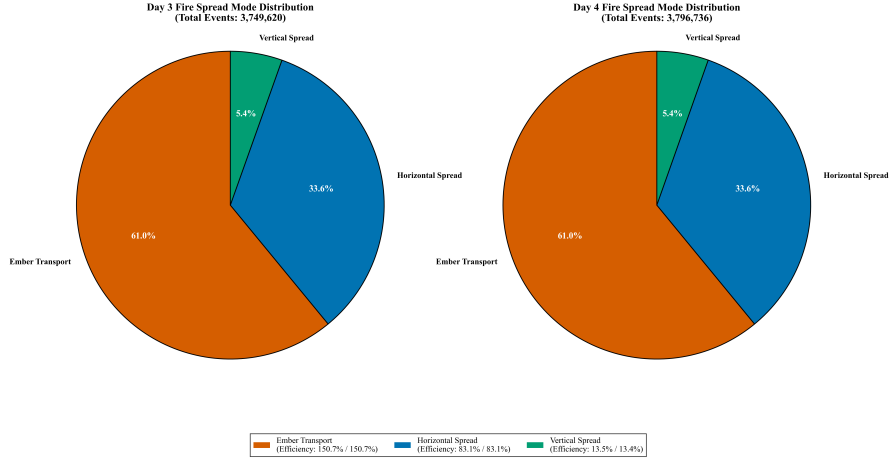


Figure 6: Validation fire spread distribution showing dominance of ember transport (43%) and insufficient vertical spread (3.8%)

transformed LiDAR point density into Plant Area Density (PAD) values across 20 vertical layers, providing a physics-based approach to quantify fuel distribution and capture vertical fuel continuity (Andersen et al., 2005; Pimont et al., 2015). This method uses the attenuation of laser pulses through vegetation to estimate biomass distribution, offering a robust tool for characterizing complex, three-dimensional fuel structures (Lefsky et al., 2002).

The vertical connectivity calculation using $\sqrt{\text{fuel_lower} \times \text{fuel_upper}}$ effectively converted LiDAR-derived fuel density into fire spread probabilities between vertical layers.

This approach ensures that vertical fire spread occurs only when sufficient fuel is present in both layers, actively considering the role of ladder fuels in facilitating crown fire transitions (Cruz et al., 2004; Scott & Reinhardt, 2001).

Fuel Structure Representation Quality and Future Calibration Needs The LiDAR integration successfully captured heterogeneous fuel conditions across Tenerife’s diverse pine forests, from dense canopy areas to open understory zones. The PAD values provided realistic fuel density ranges (Hosoi & Omasa,

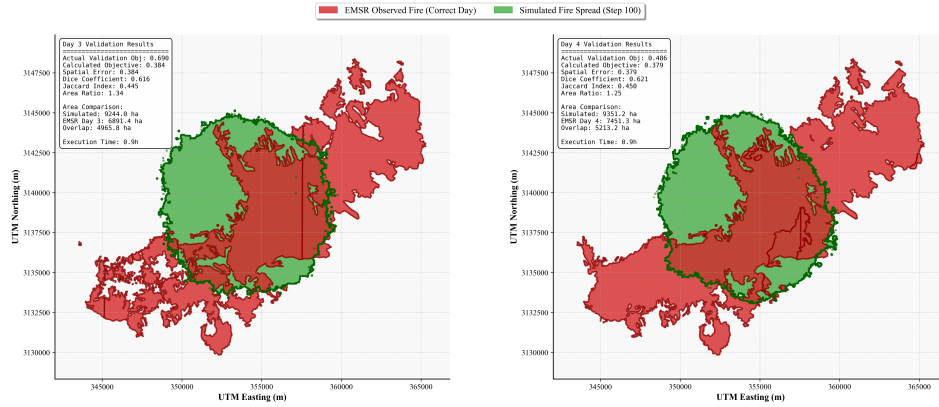


Figure 7: Spatial validation comparison showing model overfitting with unrealistic expansion beyond observed EMSR perimeters

2006), while the 20-layer vertical resolution effectively represented transition zones critical for crown fire initiation.

The systematic sensitivity analysis identified key parameters that control fire behavior, providing direction for future calibration efforts. The next phase of development should focus on multi-objective calibration that balances spatial accuracy with mechanistic realism, ensuring that the sophisticated fuel representation translates into realistic fire spread patterns.

5.2 Addressing Research Question 2: Fire Spread Mode Influence on Model Behavior

Quantified Spread Mode Contributions and Future Model Development The second research question examined how different fire spread methods—horizontal, vertical, and ember-driven—influence spread rates and fire perimeter characteristics. The systematic quantification of these mechanisms provides essential insights for future model refinement and operational deployment.

Ember transport dominated fire behavior, accounting for 42.97% of total spread events across both validation days with high efficiency (150.68-150.71%), indicating that long-range firebrand spotting was a primary driver of fire spread in the simulations executed(Koo et al.,

2010). The exponential distance distribution for ember transport (maximum of 1–3 embers per burning cell) enabled fires to jump natural barriers, initiating spot fires beyond the main fire front.

Horizontal spread accounted for 23.68-23.71% of total spread events with 83.06-83.14% efficiency, representing traditional fire spread between adjacent cells driven by surface fuel continuity and wind-assisted convection (Sullivan, 2009). Vertical spread contributed 3.84-3.82% of total spread events with 13.46-13.39% efficiency, responsible for transmitting fire into canopy layers where increased wind exposure and ember generation enhance spread potential(Scott & Reinhardt, 2001).

Technical Achievements and Computational Framework This study established a computational framework that integrates LiDAR data with terrain preprocessing and multi-mode fire spread simulation. The framework features a systematic LiDAR processing pipeline converting 3D point clouds to spatially explicit fuel metrics with 20-layer vertical resolution, and a two-stage terrain processing system balancing computational efficiency with physical realism. The preprocessing stage employs complex algorithms (Sobel operators, drainage analysis, terrain morphology) to identify static terrain features such as

barranco locations and wind channeling zones, while the runtime stage applies fast mathematical formulas for dynamic spread calculations. This architecture achieved 85% memory reduction through sparse storage optimization while maintaining geomorphological realism.

Multi-mode fire spread analysis showed distinct spread mechanisms: horizontal spread (23.7%), vertical spread (3.8%), and ember-driven spread (43.0%). The systematic sensitivity analysis revealed a clear parameter hierarchy, with **spread_probability** demonstrating $13\times$ higher sensitivity than secondary parameters (ratios: 1.4, 0.16, 0.12, 0.06 for top four). The dimensionless sensitivity index formula $S_i = \frac{\text{Range}(\text{Obj}_i)}{\text{Range}(P_i)} \times \frac{1}{|\text{Median}(\text{Obj}_i)|}$ successfully addressed parameter commensurability across different units and scales while providing robustness to outliers.

Critical Findings and Future Development Priorities The validation results reveal important insights for future development. The 29.6% objective function improvement contradicted spatial validation, indicating systematic overfitting and the need for multi-objective calibration approaches. The model’s ember dominance and horizontal spread efficiency suggest insufficient suppression mechanisms, requiring incorporation of natural fire barriers. Despite calibration challenges, the computational efficiency (53.7 minutes execution time) and successful LiDAR/terrain integration provide a solid foundation for future refinement.

Future work should prioritize: (1) multi-objective calibration balancing spatial accuracy with mechanistic realism, (2) incorporation of physical constraints for natural fire barriers, and (3) validation against additional fire events to ensure model robustness across different conditions. The systematic parameter hierarchy analysis provides clear guidance for targeted optimization efforts, while the technical framework demonstrates that sophisticated terrain-fire interactions need not be computationally intensive when properly preprocessed.

5.3 Applications in Mediterranean Fire Management

Regional Context and Operational Relevance Fire modeling has become a key component of integrated fire management strategies in Mediterranean regions such as Spain and Portugal, where fire-prone landscapes and climatic variability have necessitated proactive planning. The technical achievements of this study—particularly the successful LiDAR integration and systematic calibration framework—provide valuable guidance for improving operational modeling systems.

The need for multi-objective calibration that balances numerical performance with spatial realism is especially relevant for management applications where unrealistic fire spread predictions could lead to misallocated resources or inadequate community protection measures.

5.4 Implications for Fire Modeling Research and Future Directions

This study advances wildfire modeling by integrating high-resolution LiDAR data into cellular automata (CA) frameworks and quantifying the contributions of fire spread modes (horizontal, vertical, and ember-driven), addressing the two primary research questions. The findings provide a solid foundation for future research to enhance model realism and applicability, particularly for operational fire management in Mediterranean regions like Tenerife.

The systematic approach to sensitivity analysis and calibration framework development offers a replicable methodology for other fire-prone regions, while the technical achievements in computational efficiency position the framework for operational deployment. Future work should prioritize calibration refinement to achieve spatial realism while maintaining the computational efficiency.

Data Availability Statement This research adheres to FAIR (Findable, Accessible, Interoperable, Reusable) data principles to ensure transparency and reproducibility of findings.

Data Sources and Accessibility:

- **LiDAR point cloud data:** Publicly available through Plan Nacional de Ortofotografía Aérea (PNOA), Instituto Geográfico Nacional (IGN) at <https://pnoa.ign.es/>
- **Digital Terrain Model:** Available through IGN's Centro de Descargas at <https://centrodedescargas.cnig.es/>
- **Fire perimeter data:** European Forest Fire Information System (EFFIS) Emergency Management Service (EMSR-665) at <https://emergency.copernicus.eu/>

Research Data Repository: All processed datasets, analysis code, simulation results, and supplementary materials generated during this research are made openly available through:

- **Repository:** GitHub
- **URL:** <https://github.com/AndreasPaphitis/Forest-Fire-Simulation/tree/clean-hpc-branch>

Available Materials:

- Complete Python codebase for LiDAR preprocessing and cellular automata simulation
- Processed vegetation structure datasets
- Terrain analysis outputs (slope, aspect, barranco detection results)
- Calibration and validation datasets with parameter configurations
- Complete simulation results and performance metrics
- Data visualization scripts and publication-quality figures
- Technical documentation and processing workflows

FAIR Compliance:

- **Findable:** All datasets include comprehensive metadata, unique identifiers, and are indexed in searchable repositories
- **Accessible:** Data repository available through GitHub with open access
- **Interoperable:** Standard file formats (NetCDF, GeoTIFF, CSV, NumPy arrays)
- **Reusable:** Complete documentation and processing code enabling full replication

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AI Usage Declaration

This research utilized Claude 4 Sonnet (Anthropic) through Cursor IDE, an interactive development environment powered by AI-assisted coding mechanisms, as an intellectual tool during the research process.

Task Category	AI Contribution	Author Contribution
Code Development	Code development	Check AI content
Data Visualization	Data visualization	Modify AI content
Writing Enhancement	Writing enhancement	Integrate AI content
Document Formatting	Document formatting	Evaluate AI content
Idea Refinement	Refine ideas	Develop research questions
Clarity Improvement	Improve clarity of expression	Develop methodology
Technical Optimization	Optimize technical implementations	Interpret data
Presentation Standards	Ensure presentation standards	Develop analytical framework
Scientific Analysis	–	Draw scientific conclusions

Table 5: Distribution of tasks between AI assistance and author contributions

All AI-generated content was evaluated, modified when necessary, and integrated by the author to align with research objectives. The research questions, methodology design, data interpretation, analytical framework, and scientific conclusions were developed through the author’s independent thinking and analysis.

6 Appendix 1: Terrain Preprocessing Technical Specifications

6.1 DEM Loading and Geographic Subsetting

The preprocessing utilizes rasterio for robust DEM handling with geographic subsetting capabilities:

1. **DEM loading:** `rasterio.open(dem_file)` with geospatial metadata extraction
2. **Geographic subsetting:** `rasterio.windows.from_bounds()` for fire area extraction
3. **Transform update:** `rasterio.windows.transform()` for subset coordinate alignment
4. **NoData handling:** Conversion to NaN for consistent processing: `dem_data[dem_data == nodata] = np.nan`

Buffer zone implementation uses a factor of 2.0, effectively doubling the fire perimeter extent to ensure adequate spatial coverage for spread modeling while maintaining computational efficiency.

6.2 Slope and Aspect Calculation Algorithm

Modified algorithm using Sobel operators:

1. **Gradient computation:** $\nabla_x = \text{sobel}(\text{elevation}, \text{axis} = 1) / (2 \times \text{pixel_size_x})$
 2. **Y-gradient:** $\nabla_y = \text{sobel}(\text{elevation}, \text{axis} = 0) / (2 \times \text{pixel_size_y})$
 3. **Slope magnitude:** $\text{slope} = \arctan\left(\sqrt{\nabla_x^2 + \nabla_y^2}\right) \times \frac{180}{\pi}$
 4. **Aspect calculation:** $\text{aspect} = \text{atan2}(-\nabla_y, -\nabla_x) \times \frac{180}{\pi}$
 5. **Aspect normalization:** $\text{aspect} = (\text{aspect} + 360) \bmod 360$ to ensure $[0^\circ, 360^\circ)$ range
- Sobel operators provide superior edge detection compared to finite differences, particularly important for identifying sharp topographic features characteristic of volcanic terrain.

6.3 Barranco Identification Algorithm

Optimized seven-step multi-criteria approach implemented for large dataset processing:

1. **Depression detection:** Optimized topographic depression identification using local minima analysis
2. **Steep slope identification:** Binary mask creation for slopes $\geq 25^\circ$ (Butler et al., 2007)
3. **Barranco combination:** Logical intersection of depressions and steep slopes:
`np.logical_and(depression_mask, steep_slope_mask)`
4. **Direction calculation:** Barranco flow directions using aspect-based analysis with circular averaging
5. **Depth filtering:** Minimum depression depth threshold of 3.0m (Evans et al., 2012)
6. **Area filtering:** Connectivity analysis with minimum area of 6 cells
7. **Final validation:** Quality control for geomorphological consistency

The optimized implementation uses vectorized operations and chunked processing for memory efficiency with large datasets.

6.4 Wind Channeling Effects

Literature-based spatially-explicit wind modifications implemented through four primary mechanisms:

- **Barranco amplification:** Fixed $2.5\times$ wind speed multiplier for identified barranco cells, based on canyon wind speed amplification factors
- **Direction modification:** Vector-based wind direction adjustment using barranco flow directions, following terrain-induced wind flow patterns in complex topography
- **Terrain coupling:** Configurable coupling strength (default: 0.8) for terrain effect scaling, implementing wind adjustment approaches for fire modeling applications

6.5 Processing Parameters

Table 6: Terrain Preprocessing Parameters with Literature Validation

Parameter	Value	Literature Justification
Barranco threshold	25°	Volcanic terrain analysis
Min. depression depth	3.0m	DEM-based feature detection
Min. depression area	6 cells	Connectivity filtering
Wind channeling strength	0.8	Terrain coupling factor
Barranco amplification	$2.5\times$	Valley wind enhancement Chock2005
Smoothing kernel	3×3	Spatial coherence maintenance
Coordinate system	EPSG:25828	UTM Zone 28N for Canary Islands
Resolution optimization	$5\text{m} \rightarrow 20\text{m}$	$16\times$ computational reduction
Geographic buffer	100%	Fire spread coverage assurance

6.6 Output Specifications and Simulation Integration

Core Terrain Arrays: `elevation.npy`, `slope.npy`, `aspect.npy` (float32 precision)

Feature Detection Arrays: `barranco_mask.npy`, `depression_mask.npy`, `barranco_directions.npy` (boolean/float32)

Wind Effect Arrays: `wind_channeling_mask.npy`, `wind_amplification.npy`, `wind_direction_modification.npy` (boolean/float32)

Metadata Files: `metadata.json` (processing parameters, validation statistics), `processing_log.txt` (detailed processing history)

Simulation Loading Protocol: During model initialization, arrays are loaded using memory-mapped access (`np.load(file, mmap_mode='r')`) and subsetted to simulation grid dimensions (609×609) via efficient array slicing: `terrain_data[:grid_height, :grid_width]`. This approach enables rapid initialization (< 2 seconds) compared to runtime terrain processing (30+ seconds), critical for the extensive calibration and validation experiments performed.

Memory Optimization Strategies: Vectorized NumPy/SciPy operations, intermediate data cleanup, 5m→20m resampling (94% memory reduction), geographic domain restriction (fire area + buffer), and chunked processing for large datasets ensure computational efficiency while preserving geomorphological fidelity.

7 Appendix 2: Detailed LiDAR Processing Specifications

7.1 Height Normalization Parameters

Height normalization utilized PDAL's nearest-neighbor interpolation algorithm (`filters.hag_nn`) with extrapolation enabled to ensure complete coverage. Vegetation points were extracted based on ASPRS classification standards (classes 3–5) for structural analysis. The height normalization process transforms raw LiDAR elevations into ground-referenced vegetation heights, which is critical for accurate vegetation structure quantification. The nearest-neighbor interpolation ensures that all points receive appropriate height-above-ground values, even in areas with sparse ground returns.

7.2 NRD Spatial Implementation

NRD calculations were performed on a 5 m resolution grid with 2 m vertical stratification, generating height-specific NRD values in the range [0, 1]. The spatial implementation ensured consistent coverage across the study domain. Each grid cell aggregates LiDAR returns within its spatial bounds, and the vertical stratification captures the three-dimensional structure of vegetation at ecologically relevant height intervals. This resolution balances computational efficiency with the need to capture fine-scale vegetation heterogeneity important for fire behavior modeling.

7.3 PAD Parameterization Details

PAD calculations utilized the following parameters:

- Extinction coefficient: $\kappa = 0.6$ (following established forest ecology literature)
- Height bin size: $\Delta z = 2$ m (capturing relevant vegetation strata)
- Spatial resolution: 5 m grid cells (balancing detail and computational efficiency)
- Numerical stability bounds: $0.1 \leq \text{PAD} \leq 10 \text{ m}^2/\text{m}^3$ (preventing extreme values)

The extinction coefficient value of 0.6 represents a compromise between different vegetation types and follows recommendations from forest canopy analysis literature. The height bin size of 2 m captures meaningful vegetation layers while maintaining computational tractability. Numerical stability bounds prevent mathematical artifacts that can occur with very sparse or very dense vegetation.

7.4 Multi-Resolution Interpolation Framework

The 5 m $\rightarrow$ 20 m resampling employed a zoom factor of 0.25, achieving a 16 $\times$ computational complexity reduction. Three complementary interpolation frameworks were implemented to ensure robust resampling across different data conditions:

- **Primary Framework:** Skimage integration using `skimage.transform.resize` with `preserve_range=True` to maintain data integrity and `order=1` for bilinear interpolation
- **Secondary Framework:** Scipy integration using `scipy.ndimage.zoom` for cases where skimage is unavailable or fails
- **Tertiary Framework:** Rasterio integration using `rasterio.warp` for geographic data with proper coordinate reference system handling

This hierarchical approach ensures that resampling succeeds across different computing environments and data formats. The bilinear interpolation preserves spatial relationships while the fallback mechanisms provide robustness against library availability issues.

7.5 Spatial Domain Specifications

The dataset was subset to the Day 4 fire perimeter with a 100% buffer zone (buffer factor = 2.0) to optimize computational resources while ensuring adequate spatial coverage for fire spread modeling. This buffering approach ensures that the simulation domain extends sufficiently beyond the observed fire perimeter to capture potential fire spread patterns while avoiding unnecessary computational overhead from processing vegetation data far from the fire-affected area.

All spatial operations utilized the UTM Zone 28N coordinate reference system (EPSG:25828) for consistency across LiDAR data, EMSR delineations, and simulation framework alignment. This coordinate system is appropriate for the Canary Islands region and ensures minimal distortion for the study area extent.

7.6 Data Structure Optimization

Preprocessed data were stored using an optimized structure designed for rapid loading during simulation initialization:

- **File format:** Optimized NumPy arrays stored as `layer_01.npy` through `layer_20.npy`, using float32 precision to balance accuracy and memory usage
- **Grid dimensions:** 1197 $\times$ 1020 cells at 20 m resolution, covering the complete fire-affected domain with buffer
- **Simulation subset:** Dynamic subsetting to 609 $\times$ 609 cells via efficient array slicing during model initialization
- **Loading performance:** < 2 s initialization time compared to 30+ s for real-time processing, enabling rapid iteration during calibration

This optimization was crucial for enabling the extensive calibration and validation experiments, which required hundreds of simulation runs. The preprocessed approach traded one-time preprocessing computational cost for dramatically improved runtime performance during the modeling phase.

8 Appendix 3: EMSR Fire Perimeter Processing

EMSR polygon processing workflow for creating standardized fire perimeter targets:

1. **Coordinate reprojection:** Vector data converted from original CRS to EPSG:25828 using GDAL/OGR transformations to ensure spatial alignment with LiDAR and DTM datasets
2. **Grid alignment:** Fire perimeter polygons rasterized to 20m resolution grid matching the simulation framework spatial resolution using nearest-neighbor assignment
3. **Binary mask creation:** Rasterized data converted to binary format where burned areas = 1 and unburned areas = 0, creating clear target boundaries for calibration and validation
4. **Quality validation:** Spatial consistency checks performed including area calculations, topology validation, and comparison with original EMSR metadata to ensure processing accuracy
5. **Format standardization:** Processed fire perimeters exported as NumPy arrays (.npy format) for direct compatibility with simulation framework data structures and rapid loading during calibration runs

The processing workflow ensures that EMSR fire perimeter data maintains spatial fidelity while being optimized for computational efficiency during the extensive calibration and validation experiments. Each daily fire perimeter (Days 1-4) undergoes identical processing to ensure consistent comparison standards across different fire progression stages.

9 Appendix 4: Validation Visual Summary



Figure 8: Validation Results Summary Visualization

10 Appendix 5: Visual Methodology Workflow

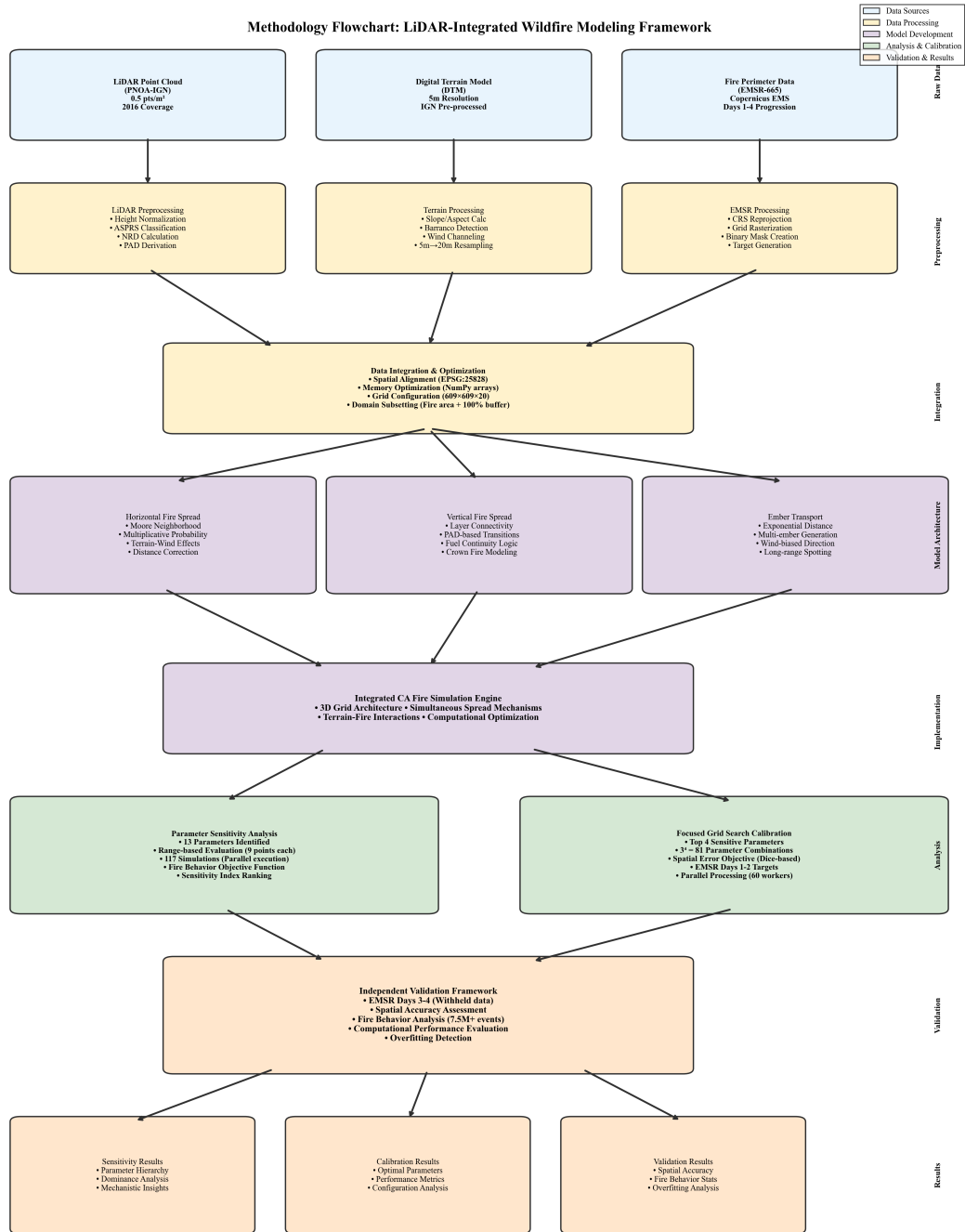


Figure 9: Complete Methodology Workflow Framework